IRRATIONALITY MEASURES FOR $L(2,\chi_{-3})$ AND OTHER DIFFERENCES OF TRIGAMMA VALUES

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ABSTRACT. Let $L = L(2,\chi_{-3})$. We prove that $\mu(aL + b\pi^2) \le 20.6622$ for all rationals a and b, not both zero. For particular directions in this plane the bound is better. Among them are

$$\mu(L) \le 13.7046, \qquad \mu\Big(L \pm \frac{\pi^2}{9}\Big) \le 9.4998, \qquad \mu\Big(L \pm \frac{2\pi^2}{27}\Big) \le 14.000.$$

The best bound known before for these numbers was 24781, due to Calegari, Dimitrov and Tang. We also prove $\mu(a\zeta(3) + b\pi^2) \le 335.20$ for all rationals a and b, not both zero. For $ab \neq 0$ these numbers are irrational by a theorem of Q. Tang, and no irrationality measure for them was known. Finally we prove

$$\mu\Big(G \pm \frac{5\pi^2}{48}\Big) \le 31.2876,$$

together with $\mu(\psi'(\frac{k}{5}) - \pi^2/6) \le 23.8272$ and $\mu(\psi'(\frac{k}{7}) - \pi^2/6) \le 1806.1719$ for k prime to the denominator. For these numbers we know of no other proof of irrationality.

The approximations are determinants of rectangular Löwner matrices of rational combinations of the digamma function and its derivative, at integer nodes. Their denominators are bounded prime by prime. For each of the two planes one table of local bounds and one decay certificate serve all directions at once. Nonvanishing at rational points follows from Hermite interpolation in characteristic zero. The size of the determinants is bounded by Andréief's identity and an energy estimate. The proofs are formalized in Lean 4.

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1. Introduction

1.1. **The results.** The values $\psi'(s) = \sum_{n\ge0}(n+s)^{-2}$ of the trigamma function at rational points $s = k/q$ are among the simplest constants of analysis. Each is a rational linear combination of π^2 and the values $L(2,\chi)$ for the Dirichlet characters χ modulo q. For $q \le 2$ they are rational multiples of π^2. Already for $q = 3$ and $q = 4$ they involve $L(2,\chi_{-3})$ and Catalan's constant $G = L(2,\chi_{-4})$, whose irrationality was proved only in 2024 and 2026 [6, 32]. Throughout the paper we work with the differences

$$\Psi(s,t) = \psi'(s) - \psi'(t) = \sum_{n=0}^{\infty}\Big(\frac{1}{(n+s)^2} - \frac{1}{(n+t)^2}\Big) \qquad (0 < s < t \le 1).$$

The *irrationality exponent* $\mu(\xi)$ of a real number ξ is the infimum of the real numbers μ such that $|\xi - a/b| > b^{-\mu}$ for all integers a and all sufficiently large integers b. It does not change under $\xi \mapsto r\xi + r'$ with $r, r' \in \mathbb{Q}$ and $r \ne 0$. A finite bound for $\mu(\xi)$ is an irrationality measure for ξ, and it shows that ξ is irrational.

We write

$$L = L(2,\chi_{-3}) = \sum_{n\ge0}\Big(\frac{1}{(3n+1)^2} - \frac{1}{(3n+2)^2}\Big) = 0.78130\,24128\ldots$$

The trigamma values with denominators 2, 3 and 6 all lie in the $\mathbb{Q}$-span of 1, L and π^2:

$$\psi'(\tfrac13) = \frac{2\pi^2}{3} + \frac{9}{2}L, \qquad \psi'(\tfrac23) = \frac{2\pi^2}{3} - \frac{9}{2}L,$$
$$\psi'(\tfrac16) = 2\pi^2 + \frac{45}{2}L, \qquad \psi'(\tfrac56) = 2\pi^2 - \frac{45}{2}L,$$
$$\psi'(\tfrac12) = \frac{\pi^2}{2}, \qquad \psi'(1) = \frac{\pi^2}{6}.$$

The first pair follows from the reflection formula $\psi'(s) + \psi'(1-s) = \pi^2/\sin^2\pi s$ and from $\psi'(\frac13) - \psi'(\frac23) = 9L$. For the second pair, $\psi'(\frac16) - \psi'(\frac56) = 36\sum_n((6n+1)^{-2} - (6n+5)^{-2})$, and splitting the series for L by residues modulo 6 shows that this sum is $\frac54 L$. So every

difference $\Psi(s,t)$ with $s,t \in \{\frac{1}{6},\frac{1}{3},\frac{1}{2},\frac{2}{3},\frac{5}{6},1\}$ is a rational point of the plane spanned by L and π^2 modulo $\mathbb{Q}$. This is our first plane.

Theorem 1.1 (first plane)**.** *Let a and b be rational numbers, not both zero. Then $aL+b\pi^2$ is irrational and*

$$\mu(aL+b\pi^2) \le 20.6622.$$

At some points of the plane we prove better bounds. These bounds do not follow from Theorem 1.1. Each has its own proof, with its own frame and geometry or with extra local bounds. The point L uses the function F_χ with two digamma terms, its own geometry, its own local bounds and its own decay certificate (Section 8). The points $L \pm \pi^2/9$ use the differences $F_{1/3}$ and $F_{2/3}$ of two digamma functions, with their own tables for Λ and their own certificate. The points in (iii)–(v) below use the frames of the plane, but with a further catalogue of local bounds that holds only at these weights. It comes from the rank of the reduction modulo p, in an assignment form (Section 7.6). We state the points in the notation Ψ, with the classical form beside it.

Theorem 1.2 (special points of the first plane)**.** *The following bounds hold.*

(i) $\mu(L) \le 13.7046$. *Here* $L = \frac{1}{9}\Psi(\frac{1}{3},\frac{2}{3})$.

(ii) $\mu(L \pm \frac{\pi^2}{9}) \le 9.4998$. *Here* $\Psi(\frac{1}{3},1) = \frac{9}{2}(L+\frac{\pi^2}{9})$ *and* $\Psi(\frac{2}{3},1) = -\frac{9}{2}(L-\frac{\pi^2}{9})$.

(iii) $\mu(\Psi(\frac{1}{6},\frac{1}{3})) \le 14.000$ *and* $\mu(\Psi(\frac{2}{3},\frac{5}{6})) \le 14.000$. *Here* $\Psi(\frac{1}{6},\frac{1}{3}) = 18(L+\frac{2\pi^2}{27})$ *and* $\Psi(\frac{2}{3},\frac{5}{6}) = 18(L-\frac{2\pi^2}{27})$.

(iv) $\mu(\Psi(\frac{1}{3},\frac{1}{2})) \le 15.359$ *and* $\mu(\Psi(\frac{1}{2},\frac{2}{3})) \le 15.359$. *Here* $\Psi(\frac{1}{3},\frac{1}{2}) = \frac{9}{2}(L+\frac{\pi^2}{27})$ *and* $\Psi(\frac{1}{2},\frac{2}{3}) = \frac{9}{2}(L-\frac{\pi^2}{27})$.

(v) $\mu(L \pm \frac{\pi^2}{18}) \le 18.161$. *Here* $L+\frac{\pi^2}{18} = \frac{1}{18}(4\Psi(\frac{1}{3},\frac{1}{2}) + \Psi(\frac{1}{2},1))$ *and* $L-\frac{\pi^2}{18} = \frac{1}{18}(4\Psi(\frac{1}{2},\frac{2}{3}) - \Psi(\frac{1}{2},1))$.

The second plane is spanned by $\zeta(3)$ and π^2 modulo $\mathbb{Q}$. It contains the values $\psi''(1) = -2\zeta(3)$ and $\psi'(1)$, $\psi'(\frac{1}{2})$, and so the numbers $\zeta(2)+\zeta(3)$ and $\zeta(3) - r\zeta(2)$ for rational r.

Theorem 1.3 (second plane)**.** *Let a and b be rational numbers, not both zero. Then*

$$\mu(a\zeta(3)+b\pi^2) \le 335.20.$$

For $ab \neq 0$ this is the first irrationality measure for these numbers. Their irrationality is due to Q. Tang, who proved in Lean that 1, $\zeta(2)$ and $\zeta(3)$ are linearly independent over $\mathbb{Q}$ [40]. At the two ends of the plane we prove the following bounds, each by its own argument. For $\zeta(3)$ the frame is $\psi'(z+1)$, with a row catalogue of its own and the decay certificate of π^2 (§ 9.6). For π^2 the function is $F_{1/2}$, with its own table for Λ. We include them for completeness. Both are weaker than the bounds $\mu(\zeta(3)) < 5.4197$ and $\mu(\pi^2) < 4.62675$ of Leibtag et al. [24].

Theorem 1.4 (the ends of the second plane)**.** *$\mu(\zeta(3)) \le 6.0382$ and $\mu(\pi^2) \le 6.5253$.*

The last three results concern numbers outside both planes. Here

$$G = L(2, \chi_{-4}) = \sum_{n \ge 0} \Big(\frac{1}{(4n+1)^2} - \frac{1}{(4n+3)^2} \Big) = 0.91596\,55941\ldots,$$

and $\psi'(\frac{1}{4}) = \pi^2 + 8G$ and $\psi'(\frac{3}{4}) = \pi^2 - 8G$, so that $\Psi(\frac{1}{4}, 1) = 8(G + \frac{5\pi^2}{48})$ and $\Psi(\frac{3}{4}, 1) = -8(G - \frac{5\pi^2}{48})$. For $q = 5$ and $q = 7$ there is no single classical constant. For $\gcd(k, q) = 1$ the number $\Psi(\frac{k}{q}, 1)$ is a rational linear combination of π^2 and the values $L(2, \chi)$ for the Dirichlet characters χ modulo q, since

$$\psi'(\tfrac{k}{q}) = q^2 \sum_{n \equiv k \ (q)} \frac{1}{n^2} = \frac{q^2}{\varphi(q)} \sum_{\chi \bmod q} \bar{\chi}(k) L(2, \chi).$$

Theorem 1.5. *The following bounds hold:*

(i) $\mu(G \pm \frac{5\pi^2}{48}) \le 31.2876$.
(ii) $\mu(\psi'(\frac{k}{5}) - \frac{\pi^2}{6}) \le 23.8272$ *for* $k = 1, 2, 3, 4$.
(iii) $\mu(\psi'(\frac{k}{7}) - \frac{\pi^2}{6}) \le 1806.1719$ *for* $k = 1, \ldots, 6$.

Each bound $\mu(\xi) \le \mu_0$ above is proved in the stronger form $|\xi - a/b| > b^{-\mu_0}$ for all integers a and all sufficiently large integers b. Table 1 collects the results.

Several numbers named in the literature lie on the first plane but at none of the points of Theorem 1.2. For them Theorem 1.1 gives $\mu \le 20.6622$ in place of 24781. These are $\psi'(\frac{1}{3}) = \frac{9}{2}(L + \frac{4\pi^2}{27})$, $\psi'(\frac{2}{3})$, $\psi'(\frac{1}{6}) = \frac{45}{2}(L + \frac{4\pi^2}{45})$ and $\psi'(\frac{5}{6})$, whose irrationality Calegari, Dimitrov and Tang state explicitly [6, Corollary B], and $\Psi(\frac{1}{6}, 1) = \psi'(\frac{1}{6}) - \zeta(2) = \frac{45}{2}(L + \frac{11\pi^2}{135})$ and $\Psi(\frac{5}{6}, 1)$. The same holds for the remaining sixths, $\Psi(\frac{1}{6}, \frac{1}{2}) = \frac{45}{2}(L + \frac{\pi^2}{15})$, $\Psi(\frac{1}{6}, \frac{2}{3}) = 27(L + \frac{4\pi^2}{81})$ and their mirrors $\Psi(\frac{1}{2}, \frac{5}{6})$, $\Psi(\frac{1}{3}, \frac{5}{6})$. Some numbers with similar names are not on the plane. The Clausen value $\mathrm{Cl}_2(\frac{\pi}{3}) = \frac{3\sqrt{3}}{4} L$ and the Mahler measure $m(1 + x + y) = \frac{3\sqrt{3}}{4\pi} L$ carry the factors $\sqrt{3}$ and $\sqrt{3}/\pi$, and our results say nothing about them. On the second plane, Theorem 1.3 gives the first measures for $\zeta(2) + \zeta(3)$ and for $\zeta(3) - r\zeta(2)$ with $r \in \mathbb{Q}^\times$. It gives none for the ratio $\zeta(3)/\pi^2$, whose irrationality also follows from Tang's theorem [23].

All the theorems above have been verified in the Lean 4 proof assistant, together with every theorem, proposition and lemma used in their proofs. The formal proofs have no unproved steps and use only the standard axioms. Section 12 describes them. The remarks and Section 11 discuss the method and are not part of the formal proofs. The formalization is available at https://github.com/slitsey/catalan-irrationality-measure. The use of AI in this work is disclosed at the end of the paper.

number	form in Ψ	$\mu \leq$	best known before
$aL + b\pi^2$, $a \neq 0$	–	20.6622	24781 [7]
L	$\frac{1}{9}\Psi(\frac{1}{3}, \frac{2}{3})$	13.7046	24781
$L \pm \frac{\pi^2}{9}$	$\Psi(\frac{1}{3}, 1)$, $\Psi(\frac{2}{3}, 1)$	9.4998	24781
$L \pm \frac{2\pi^2}{27}$	$\Psi(\frac{1}{6}, \frac{1}{3})$, $\Psi(\frac{2}{3}, \frac{5}{6})$	14.000	24781
$L \pm \frac{\pi^2}{27}$	$\Psi(\frac{1}{3}, \frac{1}{2})$, $\Psi(\frac{1}{2}, \frac{2}{3})$	15.359	24781
$L \pm \frac{\pi^2}{18}$	–	18.161	24781
$a\zeta(3) + b\pi^2$, $ab \neq 0$	–	335.20	none
$\zeta(3)$	$-\frac{1}{2}\psi''(1)$	6.0382	5.4197 [24]
π^2	$3\Psi(\frac{1}{2}, 1)$	6.5253	4.62675 [24]
$G \pm \frac{5\pi^2}{48}$	$\Psi(\frac{1}{4}, 1)$, $\Psi(\frac{3}{4}, 1)$	31.2876	none
$\psi'(\frac{k}{5}) - \frac{\pi^2}{6}$	$\Psi(\frac{k}{5}, 1)$	23.8272	none
$\psi'(\frac{k}{7}) - \frac{\pi^2}{6}$	$\Psi(\frac{k}{7}, 1)$	1806.1719	none

TABLE 1. The results of this paper. The first column is a number in common notation, the second is a form of the same number in the notation Ψ up to a rational factor, where one exists. "None" means that we know of no irrationality measure.

1.2. **Background.** *The value $L(2,\chi_{-3})$ and the first plane.* Calegari, Dimitrov and Tang [6] proved that 1, π^2 and L are linearly independent over $\mathbb{Q}$. Their proof applies a new arithmetic holonomy bound to a construction of Zagier. They list the irrationality of $\psi'(\frac{1}{3})$, $\psi'(\frac{2}{3})$, $\psi'(\frac{1}{6})$ and $\psi'(\frac{5}{6})$ as a corollary, and remark that $\psi'(\frac{n}{6})$ is irrational for every n. In [7] they made holonomy bounds effective. Theorem 5.1 there gives $\mu(L) \leq 24781$. The remark that follows it states that the same theorem and proof hold for any $\mathbb{Q}$-linear combination of L and π^2. So 24781 was the known bound at every point of the first plane. Theorem 1.1 improves it to 20.6622 everywhere, and Theorem 1.2 improves it further at five pairs of points. Leibtag et al. [24] find numerically that Apéry-like approximations from conservative matrix fields have positive approximation exponents for $L \pm \pi^2/9$. A supplement to their paper uses one such field to give an Apéry-style proof of the theorem of Calegari, Dimitrov and Tang. The exponent it proves for every point of the plane is greater than 10^{-4}.

The values $\zeta(2)$, $\zeta(3)$ and the second plane. Apéry proved that $\zeta(2)$ and $\zeta(3)$ are irrational [1], and Beukers gave his integral proofs [3]. Rhin and Viola obtained $\mu(\zeta(2)) < 5.441243$ and $\mu(\zeta(3)) < 5.513891$ with permutation groups of integrals [35, 36]. Zudilin lowered the bound for $\zeta(2)$ to 5.095412 [48]. Giraudin lowered the bounds to 5.0193784 for $\zeta(2)$ [16] and to 5.5138800 for $\zeta(3)$ [17]. Leibtag, Barkan, Kalisch, Weinbaum, Shalyt and Kaminer [24] showed that the classical constructions are one-dimensional

trajectories in conservative matrix fields. By searching over other trajectories they obtained $\mu(\pi^2) < 4.62675$ and $\mu(\zeta(3)) < 5.4197$. They also proved the uniform bound $\mu(A\zeta(3) + B\zeta(4)) < 66.3$ on the plane spanned by $\zeta(3)$ and π^4, which is the closest analogue of Theorem 1.3. The linear forms in 1, $\zeta(2)$ and $\zeta(3)$ have a longer history. Dauguet and Zudilin [10] bounded such forms from below when the coefficients satisfy divisibility and size conditions. They noted that this does not give the linear independence of 1, $\zeta(2)$, $\zeta(3)$, which remained open. For a fixed number $a\zeta(3) + b\zeta(2)$ with $ab \neq 0$ their theorem gives no measure. In 2026 Tang [40] proved the linear independence in Lean. Lai [23] then proved that 1, $\zeta(3)$ and $\zeta(4)$ are linearly independent. As far as we know, no irrationality measure for any $a\zeta(3) + b\pi^2$ with $ab \neq 0$ was known before Theorem 1.3.

Catalan's constant. In October 2026 OpenAI [32] proved that G is irrational and formalized the proof in Lean. Sun's preprint [39] had claimed a proof of the irrationality of G. Wachs [43] reports that its main estimate is not justified. Finite irrationality exponents for G were obtained after OpenAI's proof by Jha [18], Carreira [9], Bojnal [4], Xu [44] and the author [25]. Carreira [8] also records attempts to carry the determinant method to $\beta(4)$. For the arithmetic of G and related values before these results see [37, 47, 46, 21, 22] and the survey [50]. Eskandari, Murty and Nemoto [13] construct a 2-dimensional mixed motive over $\mathbb{Q}$ with period G. They use it to obtain linear forms in 1 and G. Eskandari [12] improves the exponent of rational approximations to G of Rivoal–Zudilin type from 0.52 to 0.62. These are explicit approximations, not a lower bound for all rationals. Our searches found no proof that $G - 5\pi^2/48$ or $G + 5\pi^2/48$ is irrational. This does not follow from the irrationality of G, since it is not known whether 1, G and π^2 are linearly independent over $\mathbb{Q}$.

The numbers $\psi'(\frac{k}{5}) - \pi^2/6$ *and* $\psi'(\frac{k}{7}) - \pi^2/6$. Our searches found no irrationality result for them. Prellberg [34] proved that $\mathrm{Li}_2(1/5)$ is irrational, but that is a different number.

Recent results. Since September 2026 a series of new results has appeared in this area. Several of them were obtained with AI systems. OpenAI proved $\mu(\pi) = 2$, with a Lean formalization [33]. This improves the bound $\mu(\pi) < 7.1033$ of Zeilberger and Zudilin [45]. Liu, Jiang and Zhang proved $\mu(\log r) = 2$ for every rational $r > 0$ with $r \neq 1$ [26], improving the classical bound $\mu(\log 2) < 3.5746$ of Marcovecchio [29]. Their result also covers nonzero rational combinations of such logarithms. Prellberg [34] proved that $\mathrm{Li}_2(-1/4)$, $\mathrm{Li}_2(1/5)$ and $\mathrm{Li}_2(-1/3)$ are irrational. These are three cases left open in [6]. The proof uses the integrals of Rhin–Viola and Viola–Zudilin. Fauzan [14] announced a proof that $\zeta(5)$ is irrational, using rationally normalized Hankel determinants.

Determinant methods. Irrationality proofs built on determinants of large size have a precedent in Zudilin's determinantal criterion with Hankel determinants [49]. The preprint of Liu, Jiang and Zhang on logarithms [26] uses a separated-weight interpolation determinant. The arguments of Fauzan and

Lai [14, 23] use Hankel determinants. Brown [5] proposes a multivariate transfinite-diameter test for Mellin-type integrals and applies it to π^2. The framework we use is OpenAI's [32]. In it the decay of the determinants is controlled by Cauchy determinants and Andréief's identity, and the denominators are controlled prime by prime. Our determinants are Löwner matrices, that is, matrices of divided differences. They are a matrix-valued analogue of Padé and Hermite–Padé approximation [31]. Formal proofs of irrationality results in Lean also include the irrationality of $\zeta(3)$ by Liu, Zhang and Zhi [27].

1.3. **The companion paper.** Catalan's constant is also a difference of trigamma values, $G = \frac{1}{16}\Psi(\frac{1}{4},\frac{3}{4})$, and the construction applies to this pair. For it, however, our bound for the denominators exceeds our bound for the decay, so we obtain no measure for G. The companion paper [25] refines OpenAI's determinant instead and proves $\mu(G) \le 270$. Our computations for $(\frac{1}{4},\frac{3}{4})$ are not part of the formal proof.

1.4. **What a measure requires.** Let $P(x)$ be a polynomial with rational coefficients for which $P(\xi)$ is small, and suppose that $|\xi - a/b|$ is very small. Three estimates turn this into a lower bound for $|\xi - a/b|$.

(i) *Denominators.* The coefficients of P have a common denominator of size $e^{(\Lambda+o(1))r^2}$.
(ii) *Smallness on a neighbourhood.* $|P(x)| \le e^{-(D-o(1))r^2}$ for every x with $|x-\xi| \le e^{-\rho r}$, not only at ξ.
(iii) *Nonvanishing at the point.* $P(a/b) \neq 0$ for some member of the family whose size can be chosen in terms of b.

In OpenAI's argument for G the size of the determinant is tied to a prime p, and nonvanishing is proved modulo p. Here nonvanishing is proved in characteristic zero, so the size r is free. A plane theorem needs more. A number $aL+b\pi^2$ is $\xi = F'(0)$ for a combination F of three digamma functions whose weights depend on (a,b). The three estimates must then hold with constants that do not depend on the weights. For the second plane the function F has a pole of order two, so the matrices are no longer Cauchy-like at every pole, and both the arithmetic and the analysis need a new argument.

1.5. **New ideas.** The proofs rest on the following ideas. Section 2 explains the first five in more detail.

(1) *Löwner determinants of digamma combinations.* The approximations are determinants of Löwner matrices, the matrices of divided differences, of a combination F of $\psi(z+s)$ and $\psi'(z+s)$ at integer nodes. The rows and columns share a long block of nodes, which carries the variable. A further border of row nodes and column nodes alternates beyond it. The degree in the variable stays small, while the decay is that of a larger node set.

(2) *Denominators from local bounds.* For each prime we prove a list of upper bounds for the valuation of the common denominator. Each is an explicit integer expression in a few integers attached to the prime. Integrated against $\log p$, their minimum gives the denominator exponent, and checking it reduces to finitely many comparisons of rational numbers.
(3) *Local bounds that do not see the weights.* For integer weights the polar data at a prime p depend only on the shifts. All our local bounds for the first plane read only these data. So one table of bounds serves every point of the plane. A prime dividing a weight can only lower the true valuation.
(4) *A decay certificate uniform in the weights.* The energy inequalities behind the decay estimate contain no weight. The weights enter only through a constant factor in the external field, which costs $e^{O(r)}$. One trial measure on one contour therefore proves the decay for a whole plane.
(5) *Nonvanishing in characteristic zero.* A kernel vector of the matrix at a rational point gives a solution of a Hermite interpolation problem. A discrete Wronskian bounds the corank, because one pole of the increment of F is isolated in its class modulo $\mathbb{Z}$. The bound depends only on the number of poles, counted with order.
(6) *Level returns and a gauge for the special points.* At a prime p, the partial sums of F step by the residues at its poles. For special weights this level returns to an earlier value, and whole blocks of entries become p-integral. A row operation inside each residue class modulo p, by a p-integral multiple of a common row, gives some further rows the level of the return. This turns the rank of the reduction modulo p into an assignment bound, and it gives the points of Theorem 1.2(iii)–(v).
(7) *Elimination of complete periods for a double pole.* For the second plane the Schur complement of the Löwner matrix over a union of complete periods modulo p is again of Löwner type, with generators that are constant on half-periods. This follows from the displacement identity of Löwner matrices and from $\sum_{v\in\mathbb{F}_p^\times} v^{-1} = 0$. A uniform pivot on complete double periods extends it to primes smaller than the border. The required nondegeneracy modulo p is the Wronskian argument of item 5 in characteristic p.
(8) *Jets reduced to simple poles.* At a double pole the Gram representation of the Löwner matrix carries a derivative. A rank-one splitting of this term, and Cauchy's estimate on small circles, bound every jet determinant by the Cauchy determinant of a fixed frame with simple poles. The decay certificate of that frame then applies.

The constants of the proofs are listed in Table 2.

1.6. **Organization.** Section 2 explains the construction and the main ideas without technicalities, and states the general deduction. Section 3 gives the

construction. Sections 4, 5 and 6 prove the three estimates for differences of two digamma functions. They also prove nonvanishing and decay for general combinations. Section 7 proves Theorem 1.1 and the special points of Theorem 1.2, and Section 8 treats the point L itself. Section 9 proves Theorem 1.3. Section 10 completes the proofs and gives the data. Section 11 discusses the limits of the method. Section 12 describes the formalization. The disclosure of AI use follows it.

1.7. **Notation and conventions.** We write ψ for the digamma function, the logarithmic derivative of Euler's gamma function, and v_p for the p-adic valuation. For $P \in \mathbb{Q}[x]$, $\operatorname{den} P$ is the least positive integer whose product with P lies in $\mathbb{Z}[x]$. We write $[0,r) = \{0,1,\dots,r-1\}$. For finite sets X_0, Y_0 of integers and a matrix A indexed by integers, $A[X_0,Y_0]$ is the submatrix with rows X_0 and columns Y_0. We write $\|y\|$ for the distance from a real number y to $\mathbb{Z}$, and $\|m\|_p = p\|m/p\|$.

2. Overview of the method

2.1. **Löwner matrices of digamma differences.** Let F be a function analytic near a finite set of distinct points. The *Löwner matrix* [28] of F is the matrix of divided differences indexed by the points:

$$\mathsf{L}_F[t,t'] = \frac{F(t')-F(t)}{t'-t}\ (t \neq t'), \qquad \mathsf{L}_F[t,t] = F'(t).$$

It is a Cauchy-like structured matrix. Hermite's formula gives $\mathsf{L}_F = \frac{1}{2\pi i}\oint F(z)\,v(z)v(z)^{\top}dz$ with $v_t(z) = 1/(z-t)$. A representation of F as a Stieltjes or contour integral therefore exhibits L_F as a Gram matrix of Cauchy kernels.

For a rational shift $0 < s < 1$ with denominator q we take the digamma difference F_s. For the pair $(\frac{1}{3},\frac{2}{3})$ we take a combination F_χ of three digamma functions. The nodes are integers. We put

$$\begin{gathered} F_s(z) = \psi(z+1) - \psi(z+s), \qquad \omega = \tfrac{1}{q^2}, \quad \xi_s = -\tfrac{1}{q^2}\Psi(s,1), \\ F_\chi(z) = -\tfrac{1}{3}\psi(z+\tfrac{1}{3}) + \tfrac{1}{2}\psi(z+\tfrac{1}{2}) - \tfrac{1}{6}\psi(z+1), \\ \omega = -\tfrac{1}{6}, \quad \xi_\chi = \tfrac{1}{36}\Psi(\tfrac{1}{3},\tfrac{2}{3}). \end{gathered} \tag{2.1}$$

In each case $\xi = \omega\,F'(0)$. The differences $F(t') - F(t)$ and $F'(t) - F'(0)$ are rational for integers $t, t' \geq 0$. Let R and Q be finite sets of nonnegative integer nodes, which index the rows and the columns. Then

$$\omega\,\mathsf{L}_F[R,Q] = M[R,Q] + \xi\,\mathrm{Id}_{R,Q}, \qquad \mathrm{Id}_{R,Q}[t,t'] = \mathbf{1}[t=t'], \tag{2.2}$$

with an explicit symmetric function M on $\mathbb{Z}^2_{\geq 0}$ with rational values, given in §3. Replacing ξ by a variable x gives the matrix polynomial $\Phi(x) = M[R,Q] + x\mathrm{Id}_{R,Q}$. Its determinant $P(x)$ has rational coefficients and degree at most $|R \cap Q|$, and $P(\xi) = \det(\omega\,\mathsf{L}_F[R,Q])$.

Suppose that $|\xi - a/b|$ is very small. Then $P(a/b)$ is very close to $P(\xi)$, which is tiny. On the other hand $b^{|R\cap Q|}\operatorname{den}(P)\,P(a/b)$ is an integer. If it is not zero, we get a contradiction as soon as

$$\underbrace{\log \operatorname{den} P}_{\text{denominators}} + |R \cap Q| \log b \;<\; \underbrace{-\log |P(a/b)|}_{\text{decay}}.$$

Three estimates are therefore needed: an upper bound for $\log \operatorname{den} P$, the *arithmetic estimate*; an upper bound for $|P(x)|$ near ξ, the *analytic estimate* or decay; and the *nonvanishing* of $P(a/b)$ for some member of a family of determinants whose size can be chosen in terms of b. Each estimate is proved once, for every shift, and the families differ only in a few parameters.

2.2. **Rectangular Löwner matrices with an alternating border.** We take a long common block $C = [0, r)$, which carries the variable x, and add m row nodes and m column nodes that alternate beyond it:

$$X' = \{r, r+2, \dots, r+2m-2\}, \qquad Y' = \{r+1, r+3, \dots, r+2m-1\}.$$

The rows are $R = C \sqcup X'$ and the columns are $Q = C \sqcup Y'$ (Figure 1). The determinant $P(x) = \det \Phi(x)$ has degree at most r, since the variable occurs only on the diagonal of the common block. So the factor $b^{\deg P}$ in the contradiction is that of r nodes, while the matrix has size $r + m$ and its decay is that of about $(1 + c/2)r$ nodes, where $c = 2m/r$ is the length of the border relative to r. The denominators grow too, but less, because border rows and border columns never share a node. To guarantee nonvanishing we also use the principal minors that delete a bounded number of common nodes. A larger c raises D and Λ together, and we chose c numerically to minimise $\rho/(D - \Lambda)$. Its value for each constant is given in Table 2.

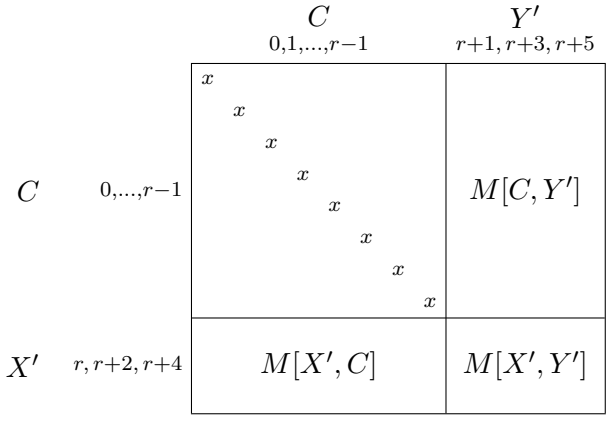


FIGURE 1. The geometry for $r = 8$ and $m = 3$. Rows are indexed by $C \sqcup X'$ and columns by $C \sqcup Y'$. The variable x occurs only on the diagonal of the common block. The border rows $X' = \{r, r+2, r+4\}$ and border columns $Y' = \{r+1, r+3, r+5\}$ alternate and never meet the diagonal.

The interleaving of X' and Y' is essential. It keeps the two Cauchy determinants in Andréief's formula comparable (§6). It also gives the reduction of the matrix modulo a prime an alternating structure. Several of the arithmetic bounds of §4 rest on this structure.

2.3. **Denominators from local bounds.** As in the work of Rhin and Viola [35, 36], we bound the p-adic valuation of the common denominator prime by prime and sum the bounds with the prime number theorem in arithmetic progressions. For $p \asymp r$ the valuation depends on r, m, p and the first pole k_p, the least index $k_p \geq 1$ at which the second digamma term has a pole modulo p. One has $k_p = \kappa p + O(1)$ with $\kappa = j/q$, where the residue $j \equiv qs\,p^{-1} \pmod q$ is the *root class* of p. The function F_χ has two first poles, and on each class of primes modulo 6 both are affine functions of one index of the same kind (Section 8).

The local bounds come from three sources: the valuations of the entries, the corank of the reduction modulo p, and cancellations at second and third order in the Smith normal form. Each is an explicit piecewise linear expression in (r, m, p, k_p), valid in a range of p/r described by linear inequalities. Replacing (r, m, p, k_p) by $(1, \alpha, \theta, \kappa\theta)$, with $\alpha = c/2$, turns it into a piecewise affine function of $\theta = p/r$, with an explicitly bounded error. The *denominator exponent* is

$$\Lambda = \frac{1}{\varphi(q)} \sum_j \int_0^\infty \mathcal{P}_j(\theta)\, d\theta,$$

where $\mathcal{P}_j$ is the minimum of these functions for the root class j. A table of affine pieces above $\mathcal{P}_j$ certifies Λ by finitely many comparisons of rational numbers (Theorem 4.14). The bounds depend on the shift only through the root class, so one table serves every shift with denominator q, and the numbers $\Psi(s, 1)$ for these shifts receive the same exponent.

2.4. **Primes smaller than the border.** For $p \leq 2m$ some residue classes modulo p contain two border nodes, a row t and a column $t + p$. The bounds above are not sharp for these primes. Each such pair gives a pivot that is a p-adic unit, because the interval $(t, t + p]$ contains one pole of each digamma term and their contributions cannot cancel. Eliminating these pivots leaves a matrix of the same shape with a border of $d = \|m\|_p$ rows and columns. To this smaller matrix we apply the separation argument used for large primes, at third order. Two identities for bordered inverses show that the elimination does not disturb the second-order terms (Lemma 4.9). A rational interpolation problem of type (n, n) shows that the remaining border nodes stay separated (Lemma 4.11).

2.5. **Nonvanishing.** Let x_0 be a rational point. The kernel of $\Phi(x_0)$ is identified with a space of pairs of polynomials that satisfy Hermite interpolation conditions at the nodes. Two such pairs have vanishing cross product, by a count of degrees and zeros. A further pair in the kernel would give a rational antidifference of the increment of F. No such antidifference exists, because

the pole at -1 is isolated in its class modulo $\mathbb{Z}$. This bounds the corank of $\Phi(x_0)$ by a number that depends only on F and on the alternation of the border. A principal restriction that deletes at most that many common nodes is then nonsingular. This is Theorem 5.1. No prime enters, so the size r can be chosen freely in terms of b.

2.6. **Decay.** Moving Hermite's contour to a polygonal path Γ in the left half-plane writes $\omega\,\mathsf{L}_F[R,Q]$ as a Gram matrix $\int v_t v_{t'}\,d\Omega$. Here Ω is a complex measure. It has signed atoms at poles of F and an exponentially small density on Γ. Andréief's identity with two families of functions, one for the rows and one for the columns, gives

$$|\det \mathsf{L}_F[R,Q]| \le \frac{1}{n_*!}\int |\mathsf{C}_R(z)|\,|\mathsf{C}_Q(z)| \prod_{i=1}^{n_*} d|\Omega|(z_i),$$

where n_* is the size of the matrix and C_R, C_Q are Cauchy determinants. Taking logarithms turns the integrand into a discrete logarithmic energy. The free points z_i interact with the nodes in an external field.

We bound this energy from below by comparing it with a *trial measure* σ, a density that is polynomial on each of finitely many segments. The comparison uses the conditional positive definiteness of the logarithmic kernel. What is left is a lower bound for one explicit function Υ on the contour, on the segment carrying the atoms and on the interval of the nodes. We prove it on a finite cover of these sets by cells, each checked in exact rational arithmetic. One theorem, Theorem 6.2, gives the decay for every geometry parameter $c \le 1$ and every function F whose poles are separated from the contour. Each family then supplies only σ, the polygon and the cover.

2.7. **The deduction.** Write $\eta = D - \Lambda$, where D is the *decay rate*, and let ρ be the *neighbourhood exponent* of the decay estimate: the decay holds at every point within $e^{-\rho r}$ of ξ. Section 10 combines the three estimates into

$$\mu(\xi) \le \frac{\rho}{\eta} = \frac{\rho}{D-\Lambda}.$$

This is the bound $d\rho/\eta$ of the companion paper [25] with degree $d = 1$ per unit of scale.

2.8. **Formal verification.** Every step is formalized in Lean 4 with Mathlib [11, 41]: the local arithmetic bounds and the conditions on the tables for Λ, the interpolation argument for nonvanishing, the contour representation, the cell checks of the decay estimate, and the prime number theorem in arithmetic progressions. The formal proofs share one core, which contains rectangular determinants, Gauss valuations, the local bounds, the tables for Λ, the selection in characteristic zero, the decay theorem and the deduction. A family supplies its function F, its geometry parameter and its numerical tables. The written proofs below are at the level of a research article. For long computations we point to the formal proof, which is the basis for the correctness of our theorems.

3. The construction

3.1. **The entries of M.** Let $0 < s < 1$ be rational with denominator q. Put $u = q(1-s)$, so that $k-1+s = (qk-u)/q$. For the function F_s of (2.1), the rational matrix M of (2.2) is as follows. Here $0 \le t < t'$ are integers, and $M[t',t] = M[t,t']$.

$$M[t,t'] = \frac{1}{q^2(t'-t)} \sum_{k=t+1}^{t'} \Big(\frac{1}{k} - \frac{q}{qk-u}\Big),$$

$$M[t,t] = -\frac{1}{q^2} \sum_{k=1}^{t} \Big(\frac{1}{k^2} - \frac{q^2}{(qk-u)^2}\Big).$$

For $s = \frac34$ this is $M[t,t'] = -\frac{1}{t'-t} \sum_{k=t+1}^{t'} \frac{1}{16k(4k-1)}$. For the function F_χ,

$$M[t,t'] = \frac{1}{6(t'-t)} \sum_{k=t+1}^{t'} g(k), \qquad M[t,t] = \frac16 \sum_{k=1}^{t} g'(k),$$

with $g(k) = \frac{1}{3k-2} - \frac{1}{2k-1} + \frac{1}{6k}$. Then $-\frac16 g'(k) = \frac{1}{2(3k-2)^2} - \frac{1}{3(2k-1)^2} + \frac{1}{36k^2}$.

We write each function as $F = \sum_{\varsigma\in\mathcal{T}} w_\varsigma \psi(z+\varsigma)$, where $\mathcal{T}$ is the set of shifts of F and $\sum_\varsigma w_\varsigma = 0$. For F_s, $\mathcal{T} = \{1, s\}$ with $w_1 = 1$ and $w_s = -1$. For F_χ, $\mathcal{T} = \{\frac13, \frac12, 1\}$. The number $|\mathcal{T}|$ of digamma terms is 2 for F_s and 3 for F_χ.

Lemma 3.1. *For each function of* (2.1), $\omega \mathsf{L}_F[R,Q] = M[R,Q] + \xi \mathrm{Id}_{R,Q}$ *for all finite sets* $R, Q \subset \mathbb{Z}_{\ge0}$, *with* ω *and* ξ *as in* (2.1).

Proof. From $\psi(w+1) - \psi(w) = 1/w$ one gets $F_s(k) - F_s(k-1) = \frac1k - \frac{1}{k-1+s} = \frac1k - \frac{q}{qk-u}$. Telescoping gives the off-diagonal entries. In the same way $\psi'(w+1) - \psi'(w) = -1/w^2$ gives $F_s'(t) - F_s'(0)$. Finally $\omega F_s'(0) = q^{-2}(\psi'(1) - \psi'(s)) = -q^{-2}\Psi(s,1) = \xi_s$. For F_χ the entries are computed in the same way. For the value at 0, the multiplication formula $\sum_{i<n} \psi'(z + \frac{i}{n}) = n^2\psi'(nz)$ gives $\psi'(\frac12) = 3\psi'(1)$ and $\psi'(\frac13) + \psi'(\frac23) = 8\psi'(1)$. Hence $F_\chi'(0) = -\frac13\psi'(\frac13) + \frac12\psi'(\frac12) - \frac16\psi'(1) = -\frac16\Psi(\frac13,\frac23)$, and $-\frac16 F_\chi'(0) = \frac{1}{36}\Psi(\frac13,\frac23) = \xi_\chi$. □

3.2. **Combinations with several terms.** The planes need combinations with more than two terms, and the second plane needs the trigamma function itself. We call a *frame* a function

$$F(z) = \sum_{s\in S} \big(a_s\psi(z+s) + b_s\psi'(z+s)\big),$$

where $S \subset \mathbb{Q} \cap (0,1]$ is a finite set of shifts, the weights a_s, b_s are rational with $(a_s, b_s) \ne (0,0)$, and $\sum_s a_s = 0$. The *order* of a shift is 1 if $b_s = 0$ and 2 otherwise. We write d for the sum of the orders, the number of poles of F in a period counted with order. The balance $\sum_s a_s = 0$ gives $F(z) = O(1/|z|)$

in the right half-plane. Put $\xi = F'(0)$ and

$$\delta(z) = F(z+1) - F(z) = \sum_{s\in S}\Big(\frac{a_s}{z+s} - \frac{b_s}{(z+s)^2}\Big),$$

and put $f_t = \sum_{k<t}\delta(k)$ and $e_t = -\sum_{k<t}\delta'(k)$. Then $f_t = F(t) - F(0)$ and $F'(t) = \xi - e_t$ for integers $t \ge 0$. So the Löwner matrix of F at nonnegative integer nodes is $\mathsf{L}_F = M + \xi\mathrm{Id}$, with $M[t,t'] = (f_{t'} - f_t)/(t'-t)$ for $t \ne t'$ and $M[t,t] = -e_t$. The function F_s of (2.1) is the frame with $S = \{s, 1\}$, $a_1 = 1$, $a_s = -1$, multiplied by ω. The geometries, the members $P_{r,I}$ and Lemma 3.5 below apply to every frame without change.

The frames of the first plane. Let $S^+ = \{\frac13, \frac12, 1\}$ and $S^- = \{\frac23, \frac12, 1\}$, and write $s_\pm$ for the first shift. For integers a, b, c with $a+b+c = 0$ and $abc \ne 0$ put

$$F^{\pm}_{a,b,c}(z) = a\psi(z+s_\pm) + b\psi(z+\tfrac12) + c\psi(z+1).$$

All shifts have order 1, so $d = 3$. The function F_χ of (2.1) is $-\frac16 F^+_{2,-3,1}$.

Lemma 3.2. *Put $\tau = (3a+2b)/(27a)$. Then $\xi = \frac{9a}{2}(L + \tau\pi^2)$ for $F^+_{a,b,c}$ and $\xi = -\frac{9a}{2}(L - \tau\pi^2)$ for $F^-_{a,b,c}$. For every rational t there is a frame on S^+ or on S^- whose number ξ is a nonzero rational multiple of $L + t\pi^2$.*

Proof. On S^+, $\xi = a\psi'(\frac13) + b\psi'(\frac12) + c\psi'(1)$. Insert the values of §1 and $c = -a-b$. The coefficient of L is $\frac{9a}{2}$ and that of π^2 is $\frac{2a}{3} + \frac{b}{2} - \frac{a+b}{6} = \frac{a}{2} + \frac{b}{3}$, which is $\frac{9a}{2}\tau$. On S^- the sign of L changes. For the second statement let $a = 2\,\mathrm{den}(t)$, $b = a(27t-3)/2$ and $c = -a-b$. These are integers with $\tau = t$. Here $a \ne 0$, $b = 0$ only for $t = \frac19$, and $c = 0$ only for $t = \frac{1}{27}$. For these two values we use S^- with the frames $(1, -3, 2)$ and $(1, -2, 1)$, for which $\tau = -\frac19$ and $\tau = -\frac{1}{27}$. □

The frames of the second plane. For integers $v \ne 0$ and u put

$$F^{(2)}_{v,u}(z) = v\,\psi'(z+1) + u\big(\psi(z+1) - \psi(z+\tfrac12)\big).$$

The shift 1 has order 2, and the shift $\frac12$ has order 1 if $u \ne 0$. So $d = 3$ for $u \ne 0$ and $d = 2$ for $u = 0$. Since $\psi''(1) = -2\zeta(3)$ and $\psi'(1) - \psi'(\frac12) = -\frac{\pi^2}{3}$,

$$\xi = -2v\,\zeta(3) - \frac{u}{3}\pi^2.$$

Given rationals $a \ne 0$ and b, choose integers $v \ne 0$ and u with $u/v = 6b/a$. Then $\xi = -\frac{2v}{a}(a\zeta(3) + b\pi^2)$. The frame $F^{(2)}_{1,0} = \psi'(z+1)$ gives $\zeta(3)$. Its increment is $\delta(k) = -(k+1)^{-2}$, so $M[t,t'] = -(t'-t)^{-1}\sum_{k=t+1}^{t'} k^{-2}$ and $M[t,t] = \sum_{k\le t} 2k^{-3}$.

Since μ is invariant under rational affine maps, the theorems of the introduction are statements about the numbers ξ of these frames and of (2.1).

3.3. **Geometries.**

Definition 3.3. Let $r, m \geq 0$. The *geometry* $\mathcal{G}(r,m)$ consists of the common nodes $C = [0,r)$, the row nodes $X' = \{r+2i : 0 \leq i < m\}$ and the column nodes $Y' = \{r+2i+1 : 0 \leq i < m\}$. Rows are indexed by $R = C \sqcup X'$ and columns by $Q = C \sqcup Y'$. Let $I \subseteq C$ be a set of deleted common nodes. The matrix

$$\Phi_{\mathcal{G},I}(x) = \left(M[t,t'] + x\,\mathbf{1}[t = t' \in C \setminus I]\right)_{t \in R\setminus I,\ t' \in Q\setminus I}$$

is square of size $|C \setminus I| + m$. Put $P_{\mathcal{G},I}(x) = \det \Phi_{\mathcal{G},I}(x) \in \mathbb{Q}[x]$. We write $\Phi_{\mathcal{G}}$ and $P_{\mathcal{G}}$ when $I = \emptyset$, and we write $\bar{N} = r + 2m$ for the number of nodes.

By Lemma 3.1, $\Phi_{\mathcal{G}}(\xi + h) = \omega \mathsf{L}_F[R,Q] + h\mathrm{Id}_{R,Q}$ for real h. The geometry with $m = 0$ is the square Löwner matrix of size r.

Definition 3.4. Let $0 < c \leq 1$ and $\alpha = c/2$. The *alternating geometry* of parameter c at an even scale r is $\mathcal{G}(r, m(r))$ with $m(r) = \lceil \alpha r - \frac{1}{2} \rceil$. Then $|m(r) - \alpha r| \leq \frac{1}{2}$, and the border $[r, r+2m)$ has length $cr + O(1)$. For a depth $K \geq 0$, its *members* are the matrices $\Phi_{r,I} = \Phi_{\mathcal{G},I}$ with $|I| \leq K$. We put $P_{r,I} = \det \Phi_{r,I}$.

For the functions F_s we use $c = \frac{3}{10}, \frac{1}{5}, \frac{3}{10}, \frac{2}{9}, \frac{1}{6}$ for $q = 2,4,3,5,7$. For the frames of the first plane we use $c = \frac{1}{6}$, for those of the second plane $c = \frac{6}{41}$, and for $\zeta(3)$ $c = \frac{3}{10}$. For F_χ we use $c = \frac{1}{2}$, as explained in Section 8. All $o(1)$ terms below are uniform over the members as $r \to \infty$.

Lemma 3.5 (integrality)**.** *$P_{\mathcal{G},I}$ has degree at most $|C \setminus I|$. If a and $b \geq 1$ are integers with $P_{\mathcal{G},I}(a/b) \neq 0$, then $|\operatorname{den}(P_{\mathcal{G},I})\, P_{\mathcal{G},I}(a/b)| \geq b^{-|C\setminus I|}$.*

Proof. The variable occurs only in the $|C \setminus I|$ diagonal entries indexed by $C \setminus I$. So each Leibniz term has degree at most $|C \setminus I|$. Then $b^{|C\setminus I|} \operatorname{den}(P_{\mathcal{G},I})\, P_{\mathcal{G},I}(a/b)$ is a nonzero integer. □

Example. Take $s = \frac{3}{4}$, $r = 2$ and $m = 1$, so that $C = \{0,1\}$, $X' = \{2\}$ and $Y' = \{3\}$. Then

$$\Phi_{\mathcal{G}}(x) = \begin{pmatrix} x & -\frac{1}{48} & -\frac{67}{7392} \\ -\frac{1}{48} & x + \frac{7}{144} & -\frac{47}{14784} \\ -\frac{17}{1344} & -\frac{1}{224} & -\frac{1}{528} \end{pmatrix}$$

and

$$P_{\mathcal{G}}(x) = -\frac{x^2}{528} - \frac{823\,x}{3725568} - \frac{239}{37158912}.$$

The variable occurs only at the common nodes 0 and 1, so the degree is $|C| = 2$ although the matrix has size 3. When $M[X', Y']$ is invertible, a Schur complement shows that $P_{\mathcal{G}}$ is, up to a constant factor, the characteristic polynomial of a rational matrix of size $|C|$, as in the square case. We do not use this factorisation.

4. Arithmetic: the denominators

In this section we bound the common denominator of the coefficients of the members $P_{r,I}$ for the functions F_s. The method is that of Rhin and Viola [35, 36]: the p-adic valuation of the denominator is bounded prime by prime, and the bounds are summed against $\log p$ with the prime number theorem in arithmetic progressions. What is new is the form of the local bounds. Each is an explicit integer expression in four integers attached to the prime, and their minimum, integrated, gives the denominator exponent.

The local bounds come from four mechanisms, of increasing depth.

(a) *Valuations of the entries.* The assignment bounds (T1)–(T5) of §4.4.1 use only the valuations of single entries and LP duality.
(b) *Corank modulo p.* The bounds (LP1), (LP2) and (R) of §§4.4.2 and 4.4.4 compute the corank of the reduction of a rescaled matrix.
(c) *Second order.* The bounds (W) and (S) of §§4.4.3 and 4.4.5 use the pairing of kernel vectors modulo p^2 (§4.3).
(d) *Third order.* The bounds (U_n) of §4.4.6 treat the primes smaller than the border.

Comparing the local bounds with a table of affine pieces is a finite computation with rational numbers, and Theorem 4.14 turns an admissible table into the arithmetic estimate. Section 8 treats the function F_χ.

4.1. **The root class of a prime.** Fix the function F_s, with s of denominator q, and the alternating geometry of parameter $c = 2\alpha$. Let $p > q$ be a prime. The indices $k \geq 1$ at which the entries of M acquire a denominator divisible by p are the multiples of p and the solutions of $qk \equiv u \pmod p$. Let k_p be the least of the latter. Then

$$qk_p = jp + u, \qquad 0 < j < q, \qquad j \equiv qs\,p^{-1} \pmod q.$$

We call j the *root class* of p and put $\kappa = \kappa_j = j/q$. Then $k_p - \kappa p = u/q \in [\frac{1}{q}, 1 - \frac{1}{q}]$. So the first pole is at distance less than one from κp, uniformly in s. Throughout we assume $q(r + 2m) < p^2$. Then every pole below $\bar{N}$ is simple, and only the first power of p matters. The primes with $p^2 \leq q(r + 2m)$ and the prime powers contribute $O(r^{3/2} \log r)$ in total.

4.2. **Gauss valuations, the assignment bound and the rank of the reduction.** Fix a prime p. Let $\mathcal{O} \subset \mathbb{Q}(x)$ be the localisation of $\mathbb{Z}_{(p)}[x]$ at the prime ideal (p). It is a discrete valuation ring with uniformiser p and residue field $\mathbb{F}_p(x)$. Its valuation v, restricted to $\mathbb{Q}[x]$, is the *Gauss valuation* $v(f) = \min_k v_p(f_k)$ for $f = \sum f_k x^k$. By Gauss's lemma $v(fg) = v(f) + v(g)$, and $v_p(\operatorname{den} f) = \max(0, -v(f))$. For a member we write $e_p(r, I) = v_p(\operatorname{den} P_{r,I})$.

Lemma 4.1 (assignment bound and LP duality)**.** *Let A be a square matrix over $\mathbb{Q}[x]$ with rows X_0 and columns Y_0. Let $W_{tt'} = \max(0, -v(A_{tt'}))$, and let $\tau(W)$ be the largest weight $\sum W_{tt'}$ of a perfect matching between X_0 and*

Y_0. *Then* $v(\det A) \geq -\tau(W)$. *If real numbers* u_t, $u'_{t'}$ *satisfy* $W_{tt'} \leq u_t + u'_{t'}$ *for all* $t \in X_0$ *and* $t' \in Y_0$, *then* $\tau(W) \leq \sum_t u_t + \sum_{t'} u'_{t'}$.

Proof. Each Leibniz term is a product of entries along a perfect matching. So its valuation is at least minus the weight of the matching. The valuation of a sum is at least the minimum of the valuations of its terms. The second statement is weak LP duality for the assignment problem. The weight of a perfect matching is at most $\sum_t u_t + \sum_{t'} u'_{t'}$. □

Lemma 4.2 (rank of the reduction)**.** *Let* (u, u') *be integer potentials with* $W_{tt'} \leq u_t + u'_{t'}$ *for the matrix* $\Phi_{r,I}$ *at* p, *of value* $\tau' = \sum_t u_t + \sum_{t'} u'_{t'}$. *Then* $\widetilde{\Phi} = \operatorname{diag}(p^u)\, \Phi_{r,I}\, \operatorname{diag}(p^{u'})$ *has entries in* $\mathcal{O}$, *and*

$$e_p(r, I) \leq \tau' - \operatorname{corank}_{\mathbb{F}_p(x)}(\widetilde{\Phi} \bmod p).$$

Proof. We have $v(\det \Phi_{r,I}) = -\tau' + v(\det \widetilde{\Phi})$. Over the discrete valuation ring $\mathcal{O}$ the Smith normal form shows $v(\det \widetilde{\Phi}) \geq \operatorname{corank}(\widetilde{\Phi} \bmod p)$. □

4.3. **Second-order information.** The rank bounds below use a refinement of Lemma 4.2 at second order. Suppose that, after a unimodular change of rows and columns over $\mathcal{O}$, the first n_1 rows and the first n_2 columns of $\widetilde{\Phi}$ are divisible by p. Suppose also that the $n_1 \times n_2$ block where they meet is divisible by p^2. Dividing the first n_1 rows by p leaves the first n_2 columns divisible by p. So

$$v(\det \widetilde{\Phi}) \geq n_1 + n_2. \tag{4.1}$$

The vectors that produce such rows and columns are kernel vectors of the reduction of $\widetilde{\Phi}$ on single residue classes. The condition on the corner is a statement about their pairing at first order, which we now describe. Let $\widetilde{\Phi} = p^2 \Phi_{r,I}$, so that all potentials equal 1. Since $q(r+2m) < p^2$, two distinct nodes $t \neq t'$ have $v_p(t'-t) \geq 1$ only if $t \equiv t' \pmod p$, and then $v_p(t'-t) = 1$. Hence $\widetilde{\Phi}$ has entries in $\mathcal{O}$. Its reduction is block diagonal over the residue classes modulo p, which we call the *fibres*. For nodes t, t' in distinct fibres,

$$\widetilde{\Phi}_{tt'} \equiv \frac{p\,(V(t') - V(t))}{t' - t} \pmod{p^2}. \tag{4.2}$$

Here $V(t) \in \mathbb{F}_p$ is the residue of p times the partial sum $\sum_{k \leq t} (\frac{1}{k} - \frac{q}{qk-u})$, up to the normalising factor q^{-2}. The function V is a step function of t which jumps only at the poles $t \equiv 0$ and $t \equiv k_p$. Since $t' - t$ is prime to p, only its residue matters in (4.2).

For a vector $\mathbf{y}$ indexed by nodes, its *moment pair* is

$$\hat{\mathbf{y}} = \Big(\sum_t y_t,\ \sum_t y_t V(t) \Big).$$

Lemma 4.3 (pairing)**.** *Let* $\mathbf{y}$ *be supported on the rows of one fibre and* $\mathbf{y}'$ *on the columns of a different fibre. Let* t *and* t' *be nodes of the first and of*

the second fibre. Then

$$\mathbf{y}^\top \widetilde{\Phi}\, \mathbf{y}' \equiv \frac{p}{t'-t} \det \begin{pmatrix} \hat{\mathbf{y}} \\ \hat{\mathbf{y}}' \end{pmatrix} \pmod{p^2}.$$

In particular the pairing vanishes modulo p^2 whenever the moment pairs of $\mathbf{y}$ and $\mathbf{y}'$ are proportional.

Proof. Expand $\mathbf{y}^\top \widetilde{\Phi}\, \mathbf{y}'$ with (4.2). □

The moment pairs are never both zero. This follows from the displacement structure of Löwner matrices.

Lemma 4.4 (moment lemma)**.** *Let A be a matrix with rows indexed by a set X_0 of distinct integers and columns by a set Y_0, with $|X_0| = |Y_0| + 1$. Suppose that its off-diagonal entries are $A(t,t') = (\mathcal{L}(t') - \mathcal{L}(t))/(t'-t)$ for some function $\mathcal{L}$, and that its diagonal entries, on $X_0 \cap Y_0$, are arbitrary. Then A has a nonzero left kernel vector $\mathbf{y}$ with $(\sum_t y_t, \sum_t y_t \mathcal{L}(t)) \neq (0,0)$.*

Proof. Write $\operatorname{diag}(X_0)$ and $\operatorname{diag}(Y_0)$ for the diagonal matrices of the indices. Then $\operatorname{diag}(X_0)A - A\operatorname{diag}(Y_0) = \mathcal{L}_{X_0}\mathbf{1}^\top - \mathbf{1}\mathcal{L}_{Y_0}^\top$, because the diagonal entries cancel on common indices. The left kernel $\mathcal{V}$ of A is nonzero. Suppose that every $\mathbf{y} \in \mathcal{V}$ had zero moment pair. Multiplying on the left by $\mathbf{y}^\top$ would give $\mathbf{y}^\top \operatorname{diag}(X_0)A = 0$. So $\operatorname{diag}(X_0)$ would preserve $\mathcal{V}$. As this matrix is diagonal with distinct entries, $\mathcal{V}$ would contain a standard basis vector. That vector is supported on one index t, and its moment pair is $(1, \mathcal{L}(t)) \neq (0,0)$. This is a contradiction. □

4.4. **The local bounds.** We now list the local bounds. Each is a statement of the following form. Let $P_{r,I}$ be a member with $|I| \leq K$, and let $p > p_0$ be a prime with $q(r+2m) < p^2$ that satisfies the stated conditions. Then

$$e_p(r, I) \leq \tau(r, m, p, k_p) + \epsilon_\tau\, |I|. \tag{4.3}$$

Here τ is the expression in the table and $\epsilon_\tau \in \{0,1\}$ is given in its last column. The constant p_0 depends only on F. We write $(\cdot)_+ = \max(\cdot, 0)$, $\|m\|_p$ for the distance of m to $p\mathbb{Z}$, and $\bar{u} = (k_p + p - r)_+$.

Theorem 4.5 (local bounds)**.** *The following bounds of the form (4.3) hold.*

	bound τ	*conditions*	ϵ_τ
(T1)	$2(r+m)$	*none*	0
(T1′)	$2r+2m-\|m\|_p$	*none*	0
(T2)	$2(r-\bar{u})_+ + 2m + \min(\bar{u}, 2m)$	$2m < p \le r$	0
(T3)	$2r+m$	$2m < p \le r$	0
(T4)	$2(r-k_p)_+ + 2m + (2m-(p-r))_+$	$r < p$	0
(T5)	$(2m-(k_p-r)_+)_+$	$\bar{N} \le p$, $r \le k_p$	0
(LP1)	$\tau_{\mathrm{T2}} - (\min(\bar{u}, 2m) - m)_+$	$p \le r < p+k_p$, $\bar{N} \le 2p$	1
(LP2)	$2(r-k_p)+3m$	$r < p < r+m$, $k_p \le r$, $\bar{N} \le 2p$	1
(W)	$2r+2m-(t_2-t_1)$ *for each window* $[t_1,t_2)$ *of Theorem 4.7*	*as there*	0
(R)	Π_R *of* (4.4)	$2m < p$	0
(S)	Π_S *of* (4.5)	$2m \le p$	1
(U_n)	$\Pi_{U,n}$ *of* (4.6), $n \ge 0$	$np \le m < (n+1)p$, *and* $p \le 2m$ *if* $n = 0$	0

The rest of this subsection proves Theorem 4.5, family by family.

4.4.1. *The assignment bounds (T1)–(T5).* These follow from Lemma 4.1 with explicit potentials. Consider an off-diagonal entry $M[t,t']$. By the ultrametric inequality, $-v_p(M[t,t'])$ is at most the number of the following two events that occur: the interval $(t,t']$ contains a pole, and p divides $t'-t$. A diagonal entry $M[t,t]$ has $-v_p \le 2$ if $t \ge k_p$ and $-v_p \le 0$ if $t < k_p$. The bound (T1) is the trivial bound, with potentials 1 on every row and column.

For (T1′) the potentials depend on the residue class modulo p. On the rows of a class they are $\frac{1}{2}$, $\frac{3}{2}$ or 1, according as X' has more, fewer or as many nodes in the class as Y'. On the columns they are $\frac{3}{2}$, $\frac{1}{2}$ or 1 in the same three cases. The border alternates between X' and Y'. So the imbalance is at most one per class, and the number of unbalanced classes is $2\|m\|_p$.

For (T2)–(T5) one starts from the potential $\mathbf{1}[t \ge k_p]$ on the common nodes t, which accounts for the diagonal. One adds 1, or $1+\mathbf{1}[t \ge p]$, on the border nodes t. Then one raises the potentials of the common nodes that have a congruent partner in the border. In each range one checks $W \le u+u'$ entrywise. Two facts are used. Two distinct nodes not congruent modulo p have weight at most 1. Two border nodes are never congruent once $2m < p$. For (T5) we have $\bar{N} \le p$ and $r \le k_p$. Then no common node accounts for the diagonal, and only the border nodes beyond k_p carry potential 1. So (T5) vanishes once $k_p \ge \bar{N}$, which happens for $p/r > \max(1+2\alpha, 1/\kappa)+o(1)$. All cases are checked in the formal proof.

4.4.2. *Primes near r: the bounds (LP1) and (LP2).* For p close to r the reduction of the rescaled matrix has a large block of rank one per residue class. Its corank can be computed. We give the proof of (LP2). The bound (LP1) applies to the primes with $p \le r < p + k_p$ and $\bar{N} \le 2p$. It is proved by the same method. With the potentials of (T2), eliminating an invertible block of common nodes brings the reduction to the form $\left(\begin{smallmatrix} \tilde{x}\mathrm{Id} & \mathcal{J}_Y \\ \mathcal{J}_X & 0 \end{smallmatrix}\right)$, where $\tilde{x}$ is the variable x plus a constant. In this form the border nodes below $p+k_p$ contribute rows and columns proportional to unit vectors. The Frobenius–König theorem bounds its rank. The saving over (T2) is $(\min(2m, p+k_p-r)-m)_+$, where $\min(2m, p+k_p-r)$ is the number of border nodes below $p+k_p$. We refer to the formal proof for the details.

Theorem 4.6. *Assume $r < p < r+m$, $k = k_p \le r$ and $\bar{N} = r+2m \le 2p$, and that every pole below $\bar{N}$ is simple. Then $v(\det \Phi_{r,\emptyset}) \ge -(2(r-k)+3m)$. The same holds for every member, with $\epsilon_\tau = 1$.*

Proof. Use the potentials $u(t) = \mathbf{1}[k \le t]$ for common nodes $t < r$ and $u(t) = 1 + \mathbf{1}[p \le t]$ for border nodes t. Their value is $\tau' = 2(r-k) + 2m + (\bar{N} - p)$. Since $\bar{N} \le 2p$, the poles below $\bar{N}$ are k, p, and $p+k$ if $p+k < \bar{N}$. Each is simple. The residues of p times the increment $\frac{1}{t} - \frac{q}{qt-u}$ at these poles are $-q/j$, 1 and $-q/(q+j)$.

Rows and columns beyond $p+k$ vanish. Let $t \in [p+k, \bar{N})$ be a border node, so that $u(t) = 2$. Let t' be a node of the other kind. If $t' < k$, then $u(t') = 0$. The partial sum over $(t', t]$ then has valuation at least -1, and $t - t' \in (p, 2p)$ is prime to p. Otherwise $u(t') \ge 1$ and $v_p(M[t', t]) \ge -2$. In both cases the rescaled entry is divisible by p.

The remaining entries. For nodes below $p+k$ the only poles are k and p. Put $C_0 = [0, k)$ and $C_1 = [k, r)$. Recall that $\omega = q^{-2}$ is the normalisation of (2.1). The reduction has the following blocks. On $C_0 \times C_0$ it is the regular part plus x on the diagonal. On $C_0 \times C_1$ the entries are $-\omega q/(j(t-t'))$ with $t' \in C_0$ and $t \in C_1$. On $C_1 \times C_1$ it is the scalar matrix $\omega(q/j)^2\mathrm{Id}$. A border node $t \in [r, p)$ against $t' \in C_0$ gives $\omega q/(j(t-t'))$. A border node $t \in [p, p+k)$ against $t' \in C_0$ gives $\omega(1 - q/j)$ if $t' = t - p$, and 0 otherwise. All other border entries vanish.

Rank. Delete the zero rows and columns found above. This does not change the rank. Eliminate the scalar block on C_1. The Schur complement on C_0 is $\tilde{x}\mathrm{Id}$ plus a constant multiple of the matrix with entries $\sum_{i=1}^{p-r} \frac{1}{(t+i)(t'+i)}$ for $t, t' \in C_0$. Here $\tilde{x}$ is x plus a constant, and the computation uses the reflection $H_{p-1-n} \equiv H_n \pmod p$ of harmonic numbers. The border nodes $t \in [r, p)$ are indexed by $i = p - t$. They are all present because $\bar{N} > p$. Even offsets are rows and odd offsets are columns, and together they cover $1 \le i \le p - r$. So row and column operations remove that multiple and leave $\left(\begin{smallmatrix} \tilde{x}\mathrm{Id} & \mathcal{J}_Y \\ \mathcal{J}_X & 0 \end{smallmatrix}\right)$. Here it is essential that x is an indeterminate, so that $\tilde{x}$ is a unit. A Schur complement on $\tilde{x}\mathrm{Id}$ then shows that the rank is $k + \mathrm{rank}(\mathcal{J}_X \mathcal{J}_Y)$. In $\mathcal{J}_X \mathcal{J}_Y$ the block between the high rows

and the high columns is zero, since their offsets $t-p$ have opposite parities. So $\operatorname{rank}(\mathcal{J}_X\mathcal{J}_Y) \le p-r$. In total the rank of the reduction is at most $(r-k)+k+(p-r)=p$, and the corank is at least $r+m-p$. Lemma 4.2 gives $v(\det) \ge (r+m-p)-\tau' = -(2(r-k)+3m)$. Deleting a common node changes the potentials by at most one. □

Under the hypothesis $\bar{N} \le p+k$ the third pole does not occur, and the proof is shorter. The border nodes beyond $p+k$ only contribute rows and columns that vanish modulo p. This observation allows the whole range $\bar{N} \le 2p$, which is automatic on the interval $r < p < r+m$.

4.4.3. *Pole-free windows: the bound (W).* Like the rank bounds, the bound (W) does not come from the valuations of the entries alone. Between two consecutive poles the step function V of (4.2) is constant. The reductions of the fibres of nearby residue classes are then translates of one another. Kernel vectors can be transported from one fibre to the next. Their pairings vanish modulo p^2 by Lemma 4.3. The corner bound (4.1) turns this into a saving at second order.

A *cut* is an integer t with $t \equiv 0$, $t \equiv k_p$ or $t \equiv r \pmod p$. A window $[t_1,t_2)$ of integers is *cut-free* if no cut lies in the open interval (t_1,t_2).

Theorem 4.7. *Assume that every pole below $\bar{N}$ is simple and $\bar{N} < p^2$. Let $n \ge 0$ with $np \le m$, and put $\bar{N}_n = r+2(m-np)$. Let $[t_1,t_2)$ be a cut-free window with $\max(r,\bar{N}_n-p) \le t_1 \le t_2 \le \min(\bar{N}_n, r+p)$. Then every member satisfies*

$$e_p(r,I) \le 2r+2m-(t_2-t_1).$$

For $n=0$ the windows lie in the border above $\max(r,\bar{N}-p)$, and the bound applies to primes $p>m$. For $n=\lfloor m/p \rfloor \ge 1$ it applies to the small primes $p \le m$. For these primes only (T1) and (T1′) are available among the assignment bounds. Read modulo p, the window range is then the arc of residues that receive an odd number of border nodes. The saving over (T1′) is the amount, if any, by which the length of the longest cut-free run of this arc exceeds $\|m\|_p$. For other n the window range is empty. The proof uses F only through its poles. For a function with several digamma terms the same statement holds when the cuts are the residues of all the poles and of r.

Proof. Let $\widetilde{\Phi} = p^2\Phi$, with independent diagonal variables on the common nodes. Every member is a coefficient of this determinant, so a bound for it bounds every member. Let $i<i'$ be two positions of the window. They lie in different residue classes. The translation $t \mapsto t+(i'-i)$ maps the nodes of the fibre of i bijectively onto those of the fibre of i'. It maps common nodes to common nodes, since no $t \equiv r$ lies in $(i,i']$. It maps border nodes to border nodes, since both classes lie in the window range. It maps X' to Y' when $i'-i$ is odd. There is no pole in $(i,i']$. So the step function V and the diagonal residues are unchanged, and so are the differences of nodes.

Hence the reduced block of the fibre of i' equals that of i when $i'-i$ is even, and its transpose when $i'-i$ is odd.

In the window the fibres alternate between one more row than column and one more column than row. Take a nonzero left kernel vector $\mathbf{y}$ of one block and transport it to every fibre of the window. It is a left kernel vector on the fibres with a surplus row and a right kernel vector on the others. Its moment pair is the same everywhere. By Lemma 4.3 all pairings between these vectors vanish modulo p^2. There are n_1 such vectors on rows and n_2 on columns, with $n_1+n_2=t_2-t_1$. Extending them to unimodular matrices, (4.1) gives $v(\det\widetilde{\Phi})\ge t_2-t_1$. Since $\det\widetilde{\Phi}=p^{2(r+m)}\det\Phi$, the theorem follows. □

4.4.4. *The rank bound (R).* Let $2m<p$. Write $r=k_0p+r_p$ with $0\le r_p<p$. Note that $k_p=\lceil jp/q\rceil$, since $0<u<q$. Put $\mathrm{arc}_0=[0,k_p-1]$ and $\mathrm{arc}_1=[k_p,p-1]$. For $k\ge 0$ and $\iota\in\{0,1\}$ let

$$n_{k,\iota}=\big|[r,r+2m)\cap[kp+\min\mathrm{arc}_\iota,\ kp+\max\mathrm{arc}_\iota]\big|,$$
$$n_{\mathrm{sq}}=\big|[\max(0,r+2m-p),\min(r,k_p))\big|,$$

and

$$\Pi_R=2(r+m)-\max\big(m+n_{\mathrm{sq}},\ n_{0,0}+n_{1,0}+2n_{\mathrm{sq}},\ n_{k_0,0},\ n_{k_0,1},\ n_{k_0+1,0},\ n_{k_0+1,1}\big). \tag{4.4}$$

Proof of the bound (R). Every node is $t=i+pk$ with residue $i\in[0,p)$ and digit k. Write $k(i)$ for the number of common nodes in the fibre of i. The border has length $2m<p$. So this fibre contains the common digits $0,\dots,k(i)-1$ and at most one border digit, namely $k(i)\in\{k_0,k_0+1\}$. This border node is a row or a column according to the parity of $i+pk(i)-r$. The step function V is constant on the fibre up to the bit $\iota(i)=\mathbf{1}[i\ge k_p]$. Thus the reduced block of a fibre depends only on $(k(i),\iota(i))$ and on the role of its border digit. Its entries have the Löwner form of Lemma 4.4.

A fibre with a border row has a left kernel vector with nonzero moment pair. A fibre with a border column has such a right kernel vector. For the digit 0 below the pole the step function vanishes. So these fibres have moment pairs proportional to $(1,0)$. A fibre of type $(1,0)$ without border digit has the zero block and contributes one row and one column. Choosing one such vector per fibre annihilates $m+n_{\mathrm{sq}}$ rows and as many columns. This gives the saving $m+n_{\mathrm{sq}}$.

Within a class of fibres of the same type (k,ι) all moment pairs are equal. So by Lemma 4.3 the pairings between the annihilated rows and columns of one class vanish modulo p^2. Then (4.1) gives a saving equal to the number of annihilated coordinates of the class. This number is $n_{k,\iota}$ for $k\ge 1$ and $(k,\iota)\ne(1,0)$. It is $n_{0,0}+n_{1,0}+2n_{\mathrm{sq}}$ for the class with moment pair $(1,0)$. Taking the best class gives (4.4). A deleted common node changes at most one fibre. Give up that fibre, and one further fibre to restore the balance of rows and columns. The trivial certificate then costs at most two annihilated

coordinates of each kind. The smaller matrix compensates for them, so $\epsilon_\tau = 0$. □

4.4.5. *Separated border nodes: the bound (S).* The bound (S) uses a different source of cancellation. Choose a window $[t_0, t_v]$ of border nodes whose residues lie in one arc between consecutive poles. Consider a border node outside it. Suppose the interval between the node and t_0 contains a singular index. Then the residues modulo p of p times the partial sums at the two nodes differ, because two singular contributions in an interval of length less than p cannot cancel. The leading coefficient in x of the bordered determinant of a fibre is this residue. So the cross product of the two bordered fibre determinants has a nonzero leading coefficient. The border node is then eliminated against t_0 with no further information. Count the residues of the window together with the residues of the eliminated border nodes. This gives an explicit bound, a minimum over the four window types $(k,\iota) \in \{k_0, k_0+1\} \times \{0,1\}$, which we now state.

Let $2m \le p$, and let k_0, r_p, arc_0, arc_1 and $n_{k,\iota}$ be as in §4.4.4. For a type (k,ι) let $\mathcal{K}_{k,\iota}$ be the interval of the integers $kp+i$ with $i \in \mathrm{arc}_\iota$, and with $i < r_p$ if $k = k_0+1$. The border nodes in $\mathcal{K}_{k,\iota}$ form the window, and there are $n_{k,\iota}$ of them. Let n_K be the number of border nodes in $\mathcal{K}_{k,\iota}$ and in its two translates $\mathcal{K}_{k,\iota} \pm p$. Put

$$n_\emptyset = p - |\mathcal{K}_{k,\iota}| - (2m - n_K), \qquad n_+ = (n_{k,\iota} - m)_+.$$

Since $2m \le p$, each residue class receives at most one border node. So n_K is the number of border nodes whose residues are those of $\mathcal{K}_{k,\iota}$. And $n_\emptyset$ is the number of the other residue classes that receive no border node. Then

$$\Pi_S = \min_{(k,\iota)} \big(2r + m - n_+ - (n_+ - n_\emptyset)_+\big), \tag{4.5}$$

where a type whose window is empty contributes $2r+m$. The bound (S) is $e_p(r,I) \le \Pi_S + |I|$. Its proof carries out the elimination above for every border node outside the window. We refer to the formal proof for it.

4.4.6. *Primes smaller than the border: the bound (U_n).* When $p \le 2m$ the border meets some residue classes more than once, and the bound (S) is not available. The bound (U_n) extends it to these primes. It works at third order in the Smith normal form. We show two things. After the elimination of a block of unit pivots, the rescaled matrix has the Smith invariants of a smaller geometry, to which the separation argument of (S) applies. The third-order information of that argument survives the elimination.

Write $m = np + \bar m$ with $0 \le \bar m < p$, so that $n = \lfloor m/p \rfloor$. Put

$$d = \|m\|_p = \min(\bar m, p - \bar m), \qquad \bar r = r + m - d.$$

For $i \in \{0,1,2\}$ and $\iota \in \{0,1\}$ let $Z_i = p\lfloor r/p \rfloor + (n+i)p$, and put

$$w_0 = Z_i + \min \mathrm{arc}_\iota, \quad t_0 = \max(\bar r, w_0), \quad t_v = \min(Z_i + \max \mathrm{arc}_\iota,\, \bar r + p - 1).$$

We call the geometry with common block $[0,\bar r)$ and border $[\bar r,\bar r+2d)$ the *auxiliary geometry* $(\bar r,d)$. In it, let $n_+=\big(\min(t_v+1,\bar r+2d)-t_0-d\big)_+$. Let n_K be the number of border nodes in $[w_0,t_v]$ and in its two translates by $\pm p$, and let $n_\emptyset=p-(t_v+1-w_0)-(2d-n_K)$. Then

$$\Pi_{U,n}=\min_{i,\iota}\Big(2(r+m)-d-n_+-(n_+-n_\emptyset)_+\Big), \tag{4.6}$$

all subtractions being truncated. This is the expression (4.5) for the auxiliary geometry $(\bar r,d)$, with the windows $[w_0,t_v]$ in place of $\mathcal K_{k,\iota}$. Note that $2(r+m)-d=2\bar r+d$.

Theorem 4.8. *Let $p>p_0$ be a prime with $q(r+2m)<p^2$, and let $n\ge 0$ with $np\le m<(n+1)p$. Assume $p\le 2m$ if $n=0$. Then every member satisfies $e_p(r,I)\le\Pi_{U,n}$.*

We first describe the proof for $n=0$, that is $m<p\le 2m$. We then give the changes for $n\ge 1$. As in the proof of Theorem 4.7 we work with $\widetilde\Phi=p^2\Phi$. We put an independent indeterminate x_t on each common diagonal entry. We work over the discrete valuation ring $\mathcal O$ of Gauss valuations in these indeterminates. Every member is a coefficient of $\det\widetilde\Phi$, up to the deletion of common rows and columns, which we treat at the end.

Pivots from pairs of border nodes. For $n=0$ there are $2m-p$ residue classes that carry two border nodes $t<t+p$, one a row and the other a column. There are $2d$ classes that carry one border node, and they lie in an interval of length $2d<p$. Eliminate the common block. Its fibres are invertible because the x_t are indeterminates. Consider a class with two border nodes. Let A be its common fibre, let $\mathbf a$ and $\mathbf a'$ be the couplings of its border column and border row to A, and let $\varkappa$ be its border entry. After the elimination the class has the pivot $\varpi=\varkappa-\mathbf a'^{\top}A^{-1}\mathbf a$. As the x_t tend to infinity, ϖ tends to a nonzero multiple of the residue of p times the increment of F over $(t,t+p]$. This interval contains exactly one pole of each digamma term. Let $k'p$ be the multiple of p in it, and t' the pole of the second term. If the two contributions cancelled, then $q(t'-k'p)\equiv u$ with $|q(t'-k'p)-u|<q$. Hence $q(t'-k'p)=u$, which is impossible for $0<u<q$. So these $2m-p$ pivots are units.

After them, the matrix of size $r+m=\bar r+d$ is equivalent over $\mathcal O$ to $\operatorname{diag}(\mathrm{Id}_{\bar r},p\,\Xi)$, with Ξ of size d. Its Schur complement on the border has the expansion

$$S=p\,S_1+p^2(S_2-\mathcal C_{UB}\mathcal B^{-1}\mathcal C_{BU})+O(p^3). \tag{4.7}$$

Here $\mathcal B$ is the diagonal of these pivots. The matrices $\mathcal C_{UB}$ and $\mathcal C_{BU}$ are the first-order couplings between these pivots and the single border nodes. The correction term in (4.7) is not negligible. The following identities account for it.

Lemma 4.9 (bordered inverses)**.** *Let A be an invertible square matrix over a commutative ring, and let $\mathbf u$ be a vector.*

(a) *Let A' be obtained from A by replacing one column by a vector $\mathbf{w}$, and suppose that A' is invertible. Let $\mathbf{u}'$ be obtained from $\mathbf{u}$ by replacing the corresponding entry by $\mathbf{u}^\top A^{-1}\mathbf{w}$. Then $\mathbf{u}'^\top A'^{-1}\mathbf{u} = \mathbf{u}^\top A^{-1}\mathbf{u}$.*

(b) *Suppose that the matrix*

$$A' = \begin{pmatrix} A & \mathbf{a} \\ \mathbf{a}'^\top & \varkappa \end{pmatrix}$$

is invertible. Then $(\mathbf{u}, y)^\top A'^{-1}(\mathbf{u}, \mathbf{a}'^\top A^{-1}\mathbf{u}) = \mathbf{u}^\top A^{-1}\mathbf{u}$ for every scalar y.

No symmetry of A or A' is assumed.

Proof. (a) Comparing columns gives $\mathbf{u}'^\top = \mathbf{u}^\top A^{-1}A'$. (b) We have $A'\,(A^{-1}\mathbf{u}, 0)^\top = (\mathbf{u}, \mathbf{a}'^\top A^{-1}\mathbf{u})^\top$. □

Moments. In the separation argument of (S), a border node is counted when its moment relative to a reference node vanishes. That moment is the bilinear form $\mathbf{u}^\top A^{-1}\mathbf{u}$ in the inverse of the common fibre A of its residue class. After the elimination, the common fibre of a class with two border nodes is replaced by a mixed block A' as in Lemma 4.9. Its lower node is a row and its upper node a column, or conversely. The product added to A'^{-1} is exactly the product subtracted in (4.7). So, by Lemma 4.9, the moments of the counted classes are unchanged. The single border nodes outside the window are separated from it as in (S), since two singular contributions in an interval of length $2d < p$ cannot cancel. Applied to the auxiliary geometry $(\bar{r}, d)$, the count of (S) then gives (4.6). It uses the third-order analogue of (4.1). Call the *coupling* of a residue class the residue modulo p of its border entry minus its moment against the reference node. Suppose that n_+ kernel vectors on each side survive, and that at most n_0 residue classes have a nonzero coupling. Then $\sum_i \min(v_i, 3) \geq d + n_+ + (n_+ - n_0)_+$, where $v_1, v_2, \ldots$ are the valuations of the Smith invariants. The first term d is the factor p of $p\,\Xi$.

Members. Deleting ℓ common nodes lowers the trivial bound $2(r+m)$ by 2ℓ. Consider a class with one deleted common node whose bordered system keeps its value at infinity ϑ (Lemma 4.10 below). Then the class keeps its moment. Indeed, let A be the common fibre and delete its last coordinate. The residual of the smaller solution is supported on that coordinate. Since the two values at infinity are equal, its pairing with the data vanishes. A class whose value at infinity changes gives a new separation instead. Classes with two or more deleted nodes, and classes outside the window, cost at most one each. Let k_1, k_2 and k' be the numbers of classes of these three kinds. Then $k_1 + 2k_2 + k' \leq \ell$. An elementary inequality for the count $n_+ + (n_+ - n_0)_+$ shows that the loss is at most 2ℓ. Hence the bound holds for every member with $\epsilon_\tau = 0$.

Larger n. Let $np \leq m < (n+1)p$. Each class of the remaining border absorbs n pairs of border nodes. Its block after the elimination is a mixed block. Its rows and columns are the nodes of $2n+1$ consecutive quotients

$\beta, \beta+1, \dots, \beta+2n$. Here β is the number of common nodes in the class, and the quotient of a node t in the class of a residue i is $(t-i)/p$. The window counts are those of the first case at the scale $r+np$, translated by np. The new difficulty is to show that a remaining border node separated from the window by a pole has a nonzero coupling. The values at infinity are now those of $(n+1)\times(n+1)$ bordered systems, which we describe first.

Lemma 4.10 (bordered systems and rational interpolation)**.** *Let $\mathbb{K}$ be a field and $\mathcal{Z}=\{0,1,\dots,2n\}$, with $(2n)!\neq 0$ in $\mathbb{K}$. Let $\mathcal{L}:\mathcal{Z}\to\mathbb{K}$, and let $\mathcal{Z}=X\sqcup Y$ with $|X|=n+1$ and $|Y|=n$. Consider the bordered system*

$$\sum_{y\in Y}\frac{\mathcal{L}(y)-\mathcal{L}(x)}{y-x}\,z_y+\vartheta=\mathcal{L}(x)\qquad(x\in X)$$

in the unknowns z_y and ϑ, with data $\mathcal{L}$. Its solutions correspond bijectively to the pairs $(\mathcal{N},\mathcal{D})$ of polynomials with $\mathcal{D}$ monic of degree n, $\deg\mathcal{N}\le n$ and $\mathcal{N}(t)=\mathcal{L}(t)\mathcal{D}(t)$ for $t\in\mathcal{Z}$. Under this correspondence ϑ is the coefficient of t^n in $\mathcal{N}$. In particular the system is nonsingular exactly when the interpolation problem of type (n,n) has a unique solution. Then $\vartheta=\lim_{t\to\infty}\mathcal{N}(t)/\mathcal{D}(t)$ depends only on $\mathcal{L}$ and not on the split. We call ϑ the value at infinity *of the system.*

Proof. Let $\mathcal{W}(t)=\prod_{y\in Y}(t-y)$, and let $\mathcal{W}_y=\mathcal{W}/((t-y)\mathcal{W}'(y))$ be the Lagrange polynomials of Y. Given a solution, put $\mathcal{D}=\mathcal{W}-\sum_y z_y\mathcal{W}'(y)\,\mathcal{W}_y$ and $\mathcal{N}=\vartheta\mathcal{W}-\sum_y z_y\mathcal{L}(y)\mathcal{W}'(y)\,\mathcal{W}_y$. Then $\mathcal{N}(y)=\mathcal{L}(y)\mathcal{D}(y)$ for $y\in Y$. Dividing $\mathcal{N}(x)-\mathcal{L}(x)\mathcal{D}(x)$ by $\mathcal{W}(x)\neq 0$ gives the equation of the row x. Conversely, Lagrange interpolation on Y recovers $z_y=-\mathcal{D}(y)/\mathcal{W}'(y)$ and ϑ from $(\mathcal{N},\mathcal{D})$. □

Lemma 4.11 (harmonic data)**.** *Let $\mathcal{A}\in\mathbb{K}$ with $\mathcal{A}+i\neq 0$ for $0\le i<2n$, and let $\mathcal{L}_{\mathcal{A}}(t)=\sum_{0\le i<t}1/(\mathcal{A}+i)$. Let the characteristic of $\mathbb{K}$ be 0 or larger than $2n$. For every split, the bordered system of Lemma 4.10 with $\mathcal{L}=\mathcal{L}_{\mathcal{A}}$ is nonsingular. Its value at infinity is $\vartheta=\mathcal{L}_{\mathcal{A}}(n)+\mathsf{c}_n$, with $\mathsf{c}_n\in\mathbb{K}$ depending only on n. Consequently, for the harmonic data $\mathcal{L}(t)=H_t$ on the quotients $\beta,\dots,\beta+2n$ the value at infinity is $H_{\beta+n}+\mathsf{c}_n$. The values at infinity of two classes with β and $\beta+1$ common nodes differ by $1/(\beta+n+1)$.*

Proof. Write Δ for the forward difference. For $l\ge 1$ one has $\Delta^l\mathcal{L}_{\mathcal{A}}(k)=(-1)^{l-1}(l-1)!/(\mathcal{A}+k)_l$, where $(y)_l$ is the rising factorial. One also has $\Delta^l[\binom{t}{k}f(t)](0)=\binom{l}{k}\Delta^{l-k}f(k)$. By Lemma 4.10 a solution is a pair $(\mathcal{N},\mathcal{D})$ with $\mathcal{N}=\mathcal{L}_{\mathcal{A}}\mathcal{D}$ on $\mathcal{Z}$. Since $\mathcal{N}$ has degree at most n, this means $\Delta^l(\mathcal{L}_{\mathcal{A}}\mathcal{D})(0)=0$ for $n<l\le 2n$. Write $\mathcal{D}=\sum_k \mathsf{d}_k\binom{t}{k}$ and insert the two formulas. After removing nonzero factors of the rows and columns, these n equations become a system for the coefficients d_k with $k<n$. Its matrix is the Cauchy matrix $\big(1/(l-k)\big)_{n<l\le 2n,\,0\le k<n}$, and its right-hand side comes from $\mathsf{d}_n=1$. The Cauchy matrix is nonsingular. So $\mathcal{D}$ is unique, which is the nonsingularity.

The same computation shows that $\mathsf{d}_k/(\mathcal{A}+k)_{n-k}$ does not depend on $\mathcal{A}$. For $k<n$, $\Delta^{n-k}\mathcal{L}_{\mathcal{A}}(k)$ is a constant multiple of $1/(\mathcal{A}+k)_{n-k}$. Hence $\vartheta = \Delta^n(\mathcal{L}_{\mathcal{A}}\mathcal{D})(0)/n! = \mathcal{L}_{\mathcal{A}}(n)+\mathsf{c}_n$ with c_n independent of $\mathcal{A}$. For the harmonic data on the quotients $\beta+t$ we have $H_{\beta+t} = H_\beta + \mathcal{L}_{\beta+1}(t)$. A constant added to $\mathcal{L}$ is added to ϑ. So the value at infinity is $H_\beta + \mathcal{L}_{\beta+1}(n) + \mathsf{c}_n = H_{\beta+n} + \mathsf{c}_n$. □

Remark 4.12. We note without proof that the interpolation problem of Lemma 4.11 can be solved in closed form. Its denominator is the polynomial $\mathcal{D}_n(t) = \sum_{k=0}^n \binom{n}{k}\binom{2n-k}{n-k}(\mathcal{A}+k)_{n-k}\,t(t-1)\cdots(t-k+1)$, of Hahn type [20]. The constant is $\mathsf{c}_n = 2H_n$. The determinant of the bordered system is $\mathrm{sf}(n)^2\,\mathrm{sf}(n-1)^2/(\prod_{x\in X,y\in Y}|x-y|\cdot\prod_{i<n}(\mathcal{A}+i)_{n+1})$, with $\mathrm{sf}(n)=\prod_{i\le n} i!$. All its prime factors are at most $2n$ or divide some $\mathcal{A}+i$. These closed forms are not used in the proof and are not part of the formalization.

Separation for $n\ge 1$. Between a remaining border node and the reference node of the window lies a cut. Suppose first that the cut is a pole of the second digamma term. Deform the shift to $s+pY$, with an indeterminate Y. Evaluate at the value of Y at which that pole becomes central. After multiplication by the vanishing factor, the bordered systems of the two classes become block systems built from Cauchy matrices on the quotients. The square blocks are nonsingular by Cauchy's determinant formula. A rectangular Cauchy matrix with $k+1$ rows and k columns has a left kernel vector with nonzero coordinate sum. Otherwise the rational function $\sum_i y_i/(T_i-T)$, formed from a kernel vector (y_i) and the row nodes T_i, would vanish at the k column nodes while its numerator has degree less than k. This gives the two values at infinity the values 0 and the weight of the term, which differ.

Suppose next that the cut is a multiple of p. Take instead the leading coefficient in Y. Every fractional pole term tends to 0, and the data become the harmonic numbers. The two classes have β and $\beta+1$ common nodes. Lemma 4.11 gives values at infinity that differ by $1/(\beta+n+1)$. This is a unit because $\beta+2n+1<p$, by $q(r+2m)<p^2$.

In both cases the comparison is made on the cleared numerators of the original determinants, not on inverses. The result returns to $Y=0$ by Gauss's lemma. The cleared denominators are products of the elementary denominators of the entries, which are units at $Y=0$. So a lower bound for the Gauss valuation of the deformed determinant gives the same lower bound for every coefficient of the original one. With these separations, the count and the treatment of the members are those of the case $n=0$.

4.5. **Profiles.** Let $\mathcal{E}$ be the set of integer expressions in r,p,m,k built from integer constants with the following operations: addition, multiplication by integer constants, truncated subtraction $(X-Y)_+$, min, max, and the distance $\|X\|_p$ of an expression X to $p\mathbb{Z}$. The quantities $p\lfloor r/p\rfloor$ and $r \bmod p$ are also allowed. Every bound of Theorem 4.5 lies in $\mathcal{E}$. For (R) and (S) this is (4.4) and (4.5). The ends of the intervals $\mathcal{K}_{k,\iota}$ are $kp = p\lfloor r/p\rfloor + (k-k_0)p$

plus 0, k_p, k_p-1, $p-1$ or r_p-1. For (U_n) this is (4.6) with $d=\|m\|_p$ and $Z_i=p\lfloor r/p\rfloor+(n+i)p$. For (W) the window ends are the cuts nearest to the ends of the window range, which are expressions of the same kind.

The *homogenisation* $\hat\tau(\theta)$ of $\tau\in\mathcal{E}$ replaces r,p,m,k by $1,\theta,\alpha,\kappa\theta$. It replaces every constant by 0, $\|X\|_p$ by $\theta\|\hat X/\theta\|$ and $p\lfloor r/p\rfloor$ by $\theta\lfloor 1/\theta\rfloor$. It is piecewise affine with rational breakpoints, finitely many in every interval bounded away from 0.

Lemma 4.13 (homogenisation)**.** *For each $\tau\in\mathcal{E}$ there is an explicit constant* $\mathrm{err}(\tau)$ *such that* $|\tau(r,p,m(r),k_p)-r\hat\tau(p/r)|\le\mathrm{err}(\tau)$ *for all* $r,p\ge1$. *It is computed recursively from* $\mathrm{err}(m)=\frac12$, $\mathrm{err}(k)=1-\frac1q$ *and the rules* $\mathrm{err}(\tau+\tau')=\mathrm{err}(\tau)+\mathrm{err}(\tau')$ *and* $\mathrm{err}(\min(\tau,\tau'))\le\max(\mathrm{err}(\tau),\mathrm{err}(\tau'))$.

Proof. The real version of τ, with the constants dropped, is positively homogeneous of degree one. Every operation is 1-Lipschitz in the maximum norm. The inputs differ from their homogeneous values by $|m-\alpha r|\le\frac12$ and $|k_p-\kappa p|\le1-\frac1q$. Induct on τ. □

The conditions of Theorem 4.5 are linear inequalities in r,p,m,k, strict or not. They homogenise to intervals of θ. At an endpoint the error of such an inequality is a combination of $m-\alpha r$ and $k_p-\kappa p$. With the rounding $m=\lceil\alpha r-\frac12\rceil$ two endpoints are *exact*. These are $\theta=2\alpha$, from $2m<p$, and $\theta=(1+2\alpha)/(1+\kappa)$, from $\bar N\le p+k$. At an exact endpoint the condition holds for every prime on the correct side. The other endpoints are exact, or hold outside a strip of primes of bounded width around the endpoint. On that strip another bound must be used.

4.6. **Tables for Λ and the arithmetic estimate.** For the root class j let $\mathcal{P}_j(\theta)$ be the minimum of the homogenised bounds $\hat\tau(\theta)$ over the bounds τ of Theorem 4.5 valid at θ. A *table for* Λ, for the function F_s and the geometry of parameter c, consists of the parameters (q,α,K) and the following data for every root class j:

- breakpoints $0<\theta_0<\theta_1<\cdots<\theta_{\mathrm{end}}$;
- on each interval $(\theta_{i-1},\theta_i]$, an affine function ℓ_i, a constant $\mathrm{err}_i\ge0$ and a local bound τ_i of Theorem 4.5, with a second bound at each inexact endpoint of the range of τ_i;
- a constant ℓ_0 for the initial interval $(0,\theta_0]$;

together with a rational number Λ_{fin}. We call the table *admissible* if it satisfies the following conditions.

(C1) For each shift of the table, every unit class modulo the modulus q^* of the prime number theorem used is assigned a root class with a table.

(C2) On each piece ℓ_i is nonnegative, and the piece lies in the validity interval of its bound. Moreover $\hat\tau$ lies below ℓ_i at both ends and at both one-sided limits of each breakpoint inside the piece.

(C3) At each inexact endpoint the second bound is valid on both sides and lies below ℓ_i there. The constant err_i exceeds the error constants of

Lemma 4.13, the numbers $\epsilon_\tau K$, the strip widths times the slopes, and the trivial bound at the scales below an explicit threshold.

(C4) $\ell_0 \geq 2+2\alpha$ and $0 < \theta_0 \leq 1$. Also $\theta_{\text{end}} \geq \max(q(1+2\alpha), p_0/2)$ and $\kappa_j \theta_{\text{end}} \geq 1+2\alpha$.

(C5) $\frac{1}{\varphi(q)} \sum_j (\ell_0\theta_0 + \sum_i \int_{\theta_{i-1}}^{\theta_i} \ell_i(\theta)\, d\theta) = \Lambda_{\text{fin}}$.

Each condition is a finite set of comparisons of rational numbers.

Theorem 4.14 (denominators)**.** *Let $0 < c \leq 1$, and let F_s be the function with shift s of denominator q. Suppose that an admissible table for Λ of depth K has s among its shifts. Then for every even r and every member with $|I| \leq K$,*

$$\log \operatorname{den} P_{r,I} \leq (\Lambda_{\text{fin}} + o(1))\, r^2.$$

Proof. By definition $\operatorname{den} P_{r,I} = \prod_p p^{e_p(r,I)}$. Fix a piece $(\theta_{i-1}, \theta_i]$ of the root class j. Let p be a prime in a unit class assigned to j, with $\theta_{i-1} r < p \leq \theta_i r$. Let r be beyond the threshold of (C3). Then the conditions of the bound τ_i hold, either exactly or outside a strip where the second bound applies. Lemma 4.13 with (C2) and (C3) gives $e_p(r,I) \leq r\,\ell_i(p/r) + \text{err}_i$. Below the threshold the trivial bound is at most err_i.

For $p \leq \theta_0 r$ the bound (T1) gives $e_p \leq \ell_0 r + O(1)$. The primes with $p^2 \leq q(r+2m)$, the prime powers and the finitely many primes below p_0 contribute $O(r^{3/2}\log r)$. For $p > \theta_{\text{end}} r$, (C4) gives $\bar{N} \leq p$ and $k_p \geq \bar{N}$, and (T5) gives $e_p = 0$.

Fix a unit class modulo q^*. By the prime number theorem in arithmetic progressions,

$$\sum_{\substack{\theta_{i-1} r < p \leq \theta_i r \\ p \text{ in the class}}} r\,\ell_i(p/r) \log p = \frac{r^2}{\varphi(q^*)} \int_{\theta_{i-1}}^{\theta_i} \ell_i(\theta)\, d\theta + o(r^2).$$

The class j receives $\varphi(q^*)/\varphi(q)$ unit classes modulo q^*, because the map that sends a unit to qs times its inverse permutes the units modulo q. Summing and using (C5) gives the theorem. The constants err_i contribute $O(r)$. □

The tables of this paper are generated from the functions $\mathcal{P}_j$ themselves. The pieces are the maximal intervals on which $\mathcal{P}_j$ is affine and attained by one bound. So $\Lambda_{\text{fin}} = \frac{1}{\varphi(q)} \sum_j \int \mathcal{P}_j$ up to the initial interval. Every bound of Theorem 4.5 is available to the tables: (T1)–(T5), (LP1), (LP2), (R), (S), the windows (W) and the bounds (U_n). The windows with $n = 0$ are used for $m < p$, between consecutive cuts.

On $(0, \alpha]$, that is for $p \leq m$, the only other bounds are (T1), (T1′) and (U_n). With (T1′) alone the profile there is the sawtooth $2 + 2\alpha - \theta\|\alpha/\theta\|$. Its integral is

$$\int_0^\alpha \Big(2 + 2\alpha - \theta\Big\|\frac{\alpha}{\theta}\Big\|\Big) d\theta = 2\alpha + \frac{\alpha^2}{2} + 2\alpha^2 \log 2,$$

since $\int_1^\infty \|y\| y^{-3}\,dy = \frac{3}{2} - 2\log 2$. The windows with $n = \lfloor m/p \rfloor \geq 1$ lower this profile on the strips $\alpha/(n+1) < \theta \leq \alpha/n$. The tables use them for $n \leq 8$, again with each window running between two consecutive cuts. For $q = 3, 4, 5, 7$ the tables also use (U_0) on $(\alpha, 2\alpha]$ and (U_n) on the strips with $1 \leq n \leq 24$. The table for $q = 2$ uses only the other bounds. Beyond the strips used, a table resolves the sawtooth on each interval $(\alpha/(n+1), \alpha/n]$ with $n < n_0$. It ends with a constant piece on $(0, \alpha/n_0]$, which costs $\alpha^2 T_{n_0}$ with $0 < T_n \leq 1/(4n^2)$. The number Λ_{fin} is then rational, and (C5) is an identity between rational numbers. The bounds depend on the shift s only through the root class, and their profiles only through (q, j, α). So one table serves all shifts with denominator q.

5. Nonvanishing

The deduction needs, for every rational x_0 and every large scale, one member of the family that does not vanish at x_0. For the functions of (2.1) this holds in characteristic zero. The number of deleted common nodes is bounded independently of r. This is the statement used in Section 10.

5.1. **Selection in characteristic zero.** Normalise F so that the divided differences of F at consecutive nodes have partial sums f_t, with $f_{t+1} - f_t = \Delta_F(t)$. The diagonal partial sums $\tilde{f}_t$ satisfy $\tilde{f}_{t+1} - \tilde{f}_t = -\Delta_F'(t)$. For F_s,

$$\Delta_F(z) = \omega\Big(\frac{1}{z+1} - \frac{1}{z+s}\Big) = \frac{\omega(s-1)}{(z+1)(z+s)}.$$

For F_χ, with $\omega = -\frac{1}{6}$,

$$\Delta_F(z) = \frac{1-z}{216\,(z+1)(z+\frac{1}{3})(z+\frac{1}{2})}.$$

Theorem 5.1 (selection)**.** *Let F be a function of (2.1), and consider the alternating geometry of parameter c. There is K, depending only on $|\mathcal{T}|$, with the following property. For every large even r and every $x_0 \in \mathbb{Q}$ some member $P_{r,I}$ with $|I| \leq K$ satisfies $P_{r,I}(x_0) \neq 0$.*

Sketch of proof. By barycentric interpolation, a vector in the right kernel of $\Phi_{r,\emptyset}(x_0)$ corresponds to a pair of polynomials (u, w) of degree less than $r + m$. The pair satisfies the *value conditions* $\mathsf{w}(t) = f_t \mathsf{u}(t)$ at the $\bar{N}$ nodes. It satisfies the *jet conditions* $\mathsf{w}'(t) = f_t \mathsf{u}'(t) + (x_0 - \tilde{f}_t)\mathsf{u}(t)$ at the common nodes t. The left kernel is described in the same way. Two such pairs have zero cross product $\mathsf{w}_1\mathsf{u}_2 - \mathsf{w}_2\mathsf{u}_1$. Indeed its degree is at most $2(r+m) - 2$, while the value and jet conditions give $\bar{N} + r = 2(r+m)$ zeros, counted with multiplicity.

Consider a pair whose degrees are small enough. Write $\Delta_F = n_\Delta / d_\Delta$ in lowest terms. The discrete Wronskian

$$\mathrm{Wr}(z) = d_\Delta(z)\big[\mathsf{w}(z+1)\mathsf{u}(z) - \mathsf{w}(z)\mathsf{u}(z+1)\big] - n_\Delta(z)\mathsf{u}(z)\mathsf{u}(z+1)$$

has more prescribed zeros at the nodes than its degree allows. So it vanishes identically. Then w/u would be a rational antidifference of Δ_F. This is impossible, because the simple pole of Δ_F at -1 is isolated in its class modulo $\mathbb{Z}$. The other poles of Δ_F are $-s$ for F_s, and $-\frac{1}{3}$ and $-\frac{1}{2}$ for F_χ. None of them is congruent to -1 modulo $\mathbb{Z}$. Clearing denominators in $\mathsf{w}(z+1)/\mathsf{u}(z+1) - \mathsf{w}(z)/\mathsf{u}(z) = \Delta_F(z)$ would propagate a root of the denominator indefinitely in one direction. Counting the zeros along chains of nodes of step 2 bounds the dimension of the kernel. So the corank of $\Phi_{r,\emptyset}(x_0)$ is at most a number K that depends only on $|\mathcal{T}|$ and on the number of such chains. That number is bounded independently of r.

Finally, the left and right kernels map to the same interpolation space. At a common node t their coordinates are the values $\mathsf{u}(t)$ times nonzero factors. No nonzero kernel pair has u vanishing at every common node. Otherwise, dividing by the node polynomial of the common nodes would give a pair of degree less than m that satisfies the value conditions at all $\bar{N}$ nodes, and the Wronskian argument applies again. Hence there is a set I of $|I| =$ corank common nodes such that no nonzero left or right kernel vector vanishes on I. The rows outside I then span the row space, and the columns outside I span the column space. So the principal minor that deletes I is nonzero. □

The theorem applies to every function of (2.1). For F_χ the pole of Δ_F at -1 is again isolated in its class modulo $\mathbb{Z}$, since the other poles are $-\frac{1}{3}$ and $-\frac{1}{2}$. In the formal proof the depth is $K \le 64$ for all the functions of this paper.

5.2. **General frames.** The argument above uses F only through its increment Δ_F and one pole of Δ_F that is isolated in its class modulo $\mathbb{Z}$. It therefore applies to every frame of §3.2, including the frames with a double pole. For a frame put

$$\mathsf{D}(z) = \prod_{s\in S} (z+s)^{w(s)}, \qquad \mathsf{A} = \mathsf{D}\,\delta,$$

where $w(s)$ is the order of s. Then D has degree d, and A is a polynomial of degree at most $d-2$, by the balance $\sum_s a_s = 0$.

Theorem 5.2 (nonvanishing for frames)**.** *Let F be a frame, and let $r \ge \max(1, d-2)$ and $m \ge 0$. For every $x_0 \in \mathbb{Q}$ some member $P_{r,I}$ with $|I| \le \lfloor d/2 \rfloor$ satisfies $P_{r,I}(x_0) \ne 0$.*

Proof. The partial sums satisfy $f_{t+1} - f_t = \mathsf{A}(t)/\mathsf{D}(t)$, and differentiating $\mathsf{D}\delta = \mathsf{A}$ gives the corresponding relation for the diagonal. Fix a shift s_0. Then $\mathsf{D}(-s_0) = 0$, and $\mathsf{A}(-s_0)$ is a_{s_0} or $-b_{s_0}$, according to the order, times $\prod_{s\ne s_0}(s-s_0)^{w(s)}$. So $\mathsf{A}(-s_0) \ne 0$. The translates $-s_0 \pm n$ with $n \ge 1$ are not roots of D, since distinct shifts in $(0,1]$ differ by less than 1. So the pole $-s_0$ of δ is isolated in its class modulo $\mathbb{Z}$, whatever its order. The proof of Theorem 5.1 then applies with $\Delta_F = \mathsf{A}/\mathsf{D}$. In it the discrete Wronskian of a kernel pair carries the factor D of degree d. Comparing its degree with its

prescribed zeros bounds the corank by $\lfloor (d+1+1-2)/2 \rfloor = \lfloor d/2 \rfloor$. The node set and the common block are each one chain of step 1, and the condition $r \geq d-2$ ensures that the common nodes see every kernel pair. The deletion follows as before. □

For the frames of the first plane $d = 3$, and one deleted node suffices. For the frames $F^{(2)}_{v,u}$ we have $\mathsf{D} = (z+1)^2(z+\frac{1}{2})$ and $\mathsf{A} = -(v+\frac{u}{2})z - \frac{v+u}{2}$ when $u \neq 0$, with $\mathsf{A}(-1) = \frac{v}{2} \neq 0$. So again one deleted node suffices. No case distinction in u is needed. The formal proof uses the same abstract interpolation argument for every frame.

6. Analytic decay

In this section we bound $|P_{r,I}(x)|$ for x near ξ. The argument is that of OpenAI [32] for square determinants, with three changes. The determinant has two families of nodes, one for the rows and one for the columns. The comparison measure is a density on finitely many segments rather than an explicit equilibrium measure. The inequalities that remain are verified on a finite cover by cells. The result is one theorem for every geometry parameter $c \leq 1$ and every function of (2.1).

6.1. **A contour representation.** Hermite's formula gives $\mathsf{L}_F[t,t'] = \frac{1}{2\pi i} \oint F(z)\upsilon_t(z)\upsilon_{t'}(z)\,dz$ with $\upsilon_t(z) = 1/(z-t)$. The contour encloses the nodes and no pole of F. Write $F = \sum_{\varsigma\in\mathcal{T}} w_\varsigma \psi(z+\varsigma)$ as in §3. The reflection formula $\psi(w) = \psi(1-w) - \pi\cot\pi w$ splits $F = \tilde{F} + E$. Here $\tilde{F}$ is analytic in the left half-plane and $O(1/|z|)$ there, and

$$E(z) = -\pi \sum_{\varsigma\in\mathcal{T}} w_\varsigma \cot\pi(z+\varsigma).$$

The function E is 1-periodic. Since the weights sum to zero, $E(z) = O(e^{-2\pi|\operatorname{Im} z|})$ for $|\operatorname{Im} z| \geq 1$. For F_s one has $E(z) = \pi[\cot\pi(z+s) - \cot\pi z]$.

Let Γ be a polygonal path, symmetric under complex conjugation. It crosses the real axis at a point $x^* < 0$ that is not a pole of F, runs through the left half-plane and ends in two rays. The poles of F are the points $-(i+\varsigma)$ with $i \geq 0$ and $\varsigma \in \mathcal{T}$. Moving the contour to Γ picks up the residues of F at the poles in $(x^*, 0)$. The part of the integrand with $\tilde{F}$ integrates to 0 over Γ. Hence, for all integers $t, t' \geq 0$,

$$\mathsf{L}_F[t,t'] = \int \upsilon_t \upsilon_{t'}\, d\Omega. \tag{6.1}$$

Here Ω is the complex measure with the mass $-\operatorname{res}_y(F)$ at each pole $y \in (x^*, 0)$ of F and the density $(2\pi i)^{-1}E(z)\,dz$ on Γ. This is a Gram representation with respect to a measure with finitely many atoms and an exponentially small density on Γ. After scaling by r, the atoms lie on a segment $[-r_0, 0]$ with density $|\mathcal{T}|$. The separation condition on x^* is a finite check on the fractional parts of $\varsigma - x^*$ for $\varsigma \in \mathcal{T}$.

6.2. **Two-family Andréief bound.** For a set $C'' \subseteq C$ of common nodes put $\mathcal{M}_{C''} = \omega\, \mathsf{L}_F[C'' \sqcup X', C'' \sqcup Y']$. Consider a member with common nodes $C' = C \setminus I$. Put $R' = C' \sqcup X'$, $Q' = C' \sqcup Y'$ and $n_* = |C'| + m$. By multilinearity in the diagonal,

$$P_{r,I}(\xi + h) = \sum_{C'' \subseteq C'} h^{|C'|-|C''|} \det \mathcal{M}_{C''}. \tag{6.2}$$

Andréief's identity [15] for the two families $(\upsilon_t)_{t\in R'}$ and $(\upsilon_{t'})_{t'\in Q'}$ gives

$$\det \mathsf{L}_F[R', Q'] = \frac{1}{n_*!} \int \det[\upsilon_t(z_i)]\, \det[\upsilon_{t'}(z_i)] \prod_{i=1}^{n_*} d\Omega(z_i).$$

Cauchy's determinant formula evaluates both factors. We take absolute values. The logarithmic potential theory behind what follows is in [38]. We obtain

$$|\det \mathsf{L}_F[R', Q']| \le \frac{1}{n_*!} \int \prod_{i<i'} |z_i - z_{i'}|^2 \prod_{\mathcal{F}\in\{R',Q'\}} \frac{\prod_{t<t'\in\mathcal{F}} |t'-t|}{\prod_{i,\, t\in\mathcal{F}} |z_i - t|} \prod_i d|\Omega|(z_i).$$

The integrand is the exponential of minus a discrete logarithmic energy. The free points z_i repel each other with weight 1. Each node family repels itself with weight $\frac{1}{2}$ and attracts the free points with weight 1. The density $|E| \approx e^{-2\pi|\operatorname{Im} z|}$ acts as an external field. The rows and columns differ only in the interleaved border. So the two node families have the same macroscopic density. After scaling by r, both are

$$\nu_c = \mathrm{Leb}|_{[0,1]} + \tfrac{1}{2}\mathrm{Leb}|_{[1,1+c]},$$

of mass $|\nu_c| = 1 + c/2$.

6.3. **The energy certificate.** For a regularisation radius $\epsilon > 0$ let $k_\epsilon(z,w) = -\frac{1}{2}\log(|z-w|^2 + \epsilon^2)$. Then $k_\epsilon(z,w) \le -\log|z-w|$.

Lemma 6.1 (conditional positive definiteness)**.** *Let $z_i \in \mathbb{C}$ be finitely many points and y_i real numbers with $\sum_i y_i = 0$. Then $\sum_{i,j} y_i y_j k_\epsilon(z_i, z_j) \ge 0$.*

Proof. We use Frullani's integral $\log T = \int_0^\infty (e^{-u} - e^{-Tu})\, du/u$ for $T > 0$. Together with $\sum_i y_i = 0$ it gives $\sum_{i,j} y_i y_j (-\log(|z_i - z_j|^2 + \epsilon^2)) = \int_0^\infty \frac{e^{-u\epsilon^2}}{u} \sum_{i,j} y_i y_j e^{-u|z_i - z_j|^2} du$. Gaussian kernels are positive definite. Indeed, expanding $e^{2u\langle z, z'\rangle}$ in powers and using the binomial theorem writes the inner sum as a convergent sum of squares. □

A *trial measure* is a probability measure σ, symmetric under conjugation, carried by finitely many segments $\mathcal{S}_i \subset \Gamma \cup [-r_0, 0] \cup [0,1]$ with rational endpoints. On each segment it has a density $\sigma_i(y)\, dy$ in the parameter $y \in [0,1]$ of the segment. Here σ_i is a polynomial with rational coefficients,

and $0 \le \sigma_i \le \bar\sigma_i$ for a rational $\bar\sigma_i$. We write $\Theta_\sigma = \sum_i \bar\sigma_i/|\mathcal{S}_i|$ for the sum of the bounds of its densities with respect to arclength.

We compare the free points and the deleted nodes with $|\nu_c|\sigma$ through Lemma 6.1. This bounds the free–free energy from below by a sum of one-point terms and the constant $|\nu_c|^2\mathcal{I}(\sigma)$. Riemann-sum comparisons replace the node sums by the potential U^{ν_c} and the node self-energy by $\mathcal{I}(\nu_c)$, up to $O(r\log r)$. With the external field one arrives at the function

$$\Upsilon(z) = |\nu_c|\,U^\sigma(z) - U^{\nu_c}(z) + (1-\varepsilon_0)\pi|\operatorname{Im} z|$$

for a small fixed $\varepsilon_0 > 0$. Here $U^\sigma(z) = \int \log\frac{1}{|z-w|}\,d\sigma(w)$ is the logarithmic potential of σ, and U^{ν_c} is that of ν_c. A *certificate* consists of σ and rational numbers Υ_Γ, Υ_0, J, $\mathcal{I}_\sigma$, $\mathcal{I}_\nu$ with the following properties.

(a) $\Upsilon \ge \Upsilon_\Gamma$ on Γ.
(b) $\Upsilon \ge \Upsilon_0$ on $[0,1]$, the scaled position of the deleted common nodes.
(c) $J \le \int_{-r_0}^{0} \min(\Upsilon(x) - \Upsilon_\Gamma, 0)\,dx$.
(d) $\mathcal{I}(\sigma) \le \mathcal{I}_\sigma$ and $\mathcal{I}(\nu_c) \ge \mathcal{I}_\nu$.

Theorem 6.2 (decay on a neighbourhood)**.** *Let $0 < c \le 1$. Let F be a function of (2.1) whose poles are separated from x^*. Let $(\sigma, \Upsilon_\Gamma, \Upsilon_0, J, \mathcal{I}_\sigma, \mathcal{I}_\nu)$ be a certificate for Γ and ν_c. Let $\epsilon > 0$, $\delta_\epsilon \ge \pi\epsilon\,\Theta_\sigma$, $\delta_{\mathrm{m}} > 0$ and $\delta_J > 0$ be rational, and put*

$$\delta = 3|\nu_c|^2\delta_\epsilon + 4|\nu_c|^2\delta_{\mathrm{m}} + 2|\mathcal{T}|\,\delta_J.$$

Let D and ρ be rational numbers with

$$D \le 2(|\nu_c|\Upsilon_\Gamma + |\mathcal{T}|J) - |\nu_c|^2\mathcal{I}_\sigma + \mathcal{I}_\nu - \delta, \qquad \rho \ge 2(\Upsilon_\Gamma - \Upsilon_0).$$

Then for every depth K, every member of the alternating geometry of parameter c and every real h with $|h| \le e^{-\rho r}$,

$$|P_{r,I}(\xi + h)| \le \exp\bigl(-(D - o(1))r^2\bigr).$$

The loss δ comes from replacing σ by finitely many atoms in the comparison with the free points. The regularisation radius ϵ of Lemma 6.1 and the mesh of the atoms lower Υ_Γ by $|\nu_c|(\delta_\epsilon + \delta_{\mathrm{m}})$. They raise $\mathcal{I}_\sigma$ by $\delta_\epsilon + 2\delta_{\mathrm{m}}$. Darboux sums for the integral in (c) lower J by δ_J. Inserted in the expression for D, these changes give δ. The tolerances δ_{m} and δ_J may be any positive rational numbers, since the atoms can be taken as fine as needed. In the certificates of this paper $\epsilon = 10^{-12}$ and $\delta < 10^{-9}$.

The node energy $\mathcal{I}(\nu_c)$ is a finite rational combination of the values $\mathcal{H}(y_k)$, where $\mathcal{H}(y) = \frac{1}{2}y^2\log|y| - \frac{3}{4}y^2$ and the y_k are the rational differences of the endpoints of the two pieces of ν_c. The lower bound $\mathcal{I}_\nu$ is checked with rational enclosures of the logarithms. These come from the series of artanh and a reduction to $[1,2)$. The energy $\mathcal{I}(\sigma)$ is a finite sum of pair energies of polynomial densities on segments. Each is a closed form of the type polynomial times logarithm plus arctangent, and is bounded in the same way. The conditions (a)–(c) are statements about Υ on sets of positive measure. They are proved on a cover by cells.

6.4. **The cover by cells.** Each of the conditions (a)–(c) is checked on a finite cover by cells with exact rational arithmetic. On a cell, the potential U^{σ} is split into the contribution of the segments of σ near the cell and the rest. A near segment is bounded by a Taylor polynomial in the cell variable with an explicit remainder. For a cell that meets the line of a segment beyond its end, the same bound is used for the segment with its orientation reversed. At a corner of Γ the two incident segments are treated together. Far segments are bounded by a multipole expansion with an explicit tail. The potential U^{ν_c} is an explicit elementary function, and it is bounded above by monotonicity on the cell.

Some cells end at a breakpoint of σ, that is, at an endpoint of one of its segments or at a corner of Γ. Let s be the distance from the breakpoint along the cell, with $0 < s \le S$. The potential of the segments that meet at the breakpoint is then a polynomial in s plus a term $\sum_m b_m s^m \log s$, up to an explicit remainder. In most rows the logarithmic part is bounded in absolute value by $\sum_m |b_m| S^m |\log S|$. For the cells on Γ of the shifts $\frac{k}{7}$ we use a sharper bound. For $0 < t \le 1$ and an order $K \ge 0$ write

$$-\log t = T_K(t) + R_K(t), \qquad T_K(t) = \sum_{k=1}^{K} \frac{(1-t)^k}{k}.$$

Then $0 \le R_K(t) \le (1-t)^{K+1}/((K+1)t)$, and the weighted inequality of arithmetic and geometric means gives

$$0 \le t^m R_K(t) \le \frac{(m-1)^{m-1}(K+1)^K}{(m+K)^{m+K}} \qquad (m \ge 1),$$

with $0^0 = 1$. We put $t = s/S$ and write $\log s = \log S + \log t$. This bounds each term $b_m s^m \log s$ from below by a polynomial of degree $m+K$ in s, minus a constant. Here $\log S$ is replaced by a rational number within 10^{-11} of it, checked with the enclosures of the logarithm described above. The condition on the cell becomes an inequality for one polynomial.

For (a) the cells are intervals of the arclength on each edge of Γ. The bound is proved on a horizontal neighbourhood of the edge of fixed width. Beyond the last vertex the ray is handled by an analytic tail bound. For (b) the cells are intervals of $[0,1]$. For (c) the integrand is bounded above on each cell of $[-r_0, 0]$ by a constant or by a chord. Then J is the sum of the integrals of these majorants.

Proposition 6.3 (certified constants)**.** *For each row of Table 2 there is a trial measure σ on a polygon Γ, with a cover by cells satisfying the checks above, for which Theorem 6.2 gives the constants D and ρ of the table. For the rows of the second plane the certificate is that of the reference frame of §6.6.*

All inequalities are verified by the Lean kernel, as described in Section 12. The certificate data are the polygon, the density σ and the cells. They were computed with floating-point optimisation and interval arithmetic in Arb

[19]. Lean checks them with rational arithmetic only. The data play no role in the proof beyond satisfying these checks.

Remark 6.4 (how good a certificate can be). For an exponent μ_0 the certificate meets μ_0 exactly when $\mu_0(D-\Lambda)-\rho\ge 0$. This quantity is a concave function of σ. Its maximum over all probability measures on $\Gamma\cup[-r_0,0]\cup[0,1]$ is attained at a measure σ^*. For this measure $|\nu_c|\sigma^*$ is the weighted equilibrium measure of the constrained energy problem on this set, with the external field $(1-\varepsilon_0)\pi|\operatorname{Im} z|-U^{\nu_c}$ and the upper constraint $|\mathcal{T}|\,dx$ on the segment [38]. Weak duality expresses the deficit of any certificate as a sum of nonnegative terms. The first is the energy of $2\mu_0|\nu_c|(\sigma-\sigma^*)$. This is the difference between $2\mu_0|\nu_c|\sigma$ and the optimal dual measure $2\mu_0|\nu_c|\sigma^*$. So a certificate close to the best possible one must be close to σ^* in energy. Then $\Upsilon-\Upsilon_\Gamma$ is a small ripple on the support of σ^*. This is why a pointwise verification on a cover is needed. This remark is not used in any proof and is not part of the formalization.

6.5. **Node removal and the neighbourhood.** The argument applies to every subset $C''\subseteq C'$ of the common nodes. Removing common nodes turns them into points of the configuration that are not integrated over. Only common nodes can be removed. A node removed from one family only would leave an uncancelled term of weight $\frac12$. By condition (b) one obtains

$$|\det \mathcal{M}_{C''}|\le \exp\bigl(-(D-o(1))r^2+\rho r\,|C'\setminus C''|\bigr),$$

uniformly in C''. Insert this into (6.2) with $|h|e^{\rho r}\le 1$. The sum has at most $2^{|C'|}=e^{O(r)}$ terms, each at most $e^{-(D-o(1))r^2}$. This gives Theorem 6.2. The deletion of at most K common nodes in a member changes the energy by $O(r)$.

6.6. **Frames with several terms and double poles.** For a frame of §3.2 with all orders 1 the argument above applies verbatim, with $|\mathcal{T}|=d$. It reads the frame only through the source bound $|E(z)|\le C_E e^{-2\pi|\operatorname{Im} z|}$ on Γ, through the positions of the poles and through the bound on the atom masses. The weights enter only through C_E and the atom masses. A change of these constants multiplies each term of Andréief's integral by at most $e^{O(r)}$, which is $e^{o(r^2)}$. So a certificate for a set of shifts serves every frame on those shifts. We use this for the first plane in Section 7.

At a double pole the Gram representation carries a derivative. Write

$$E(z)=-\pi\sum_s a_s\cot\pi(z+s)+\pi^2\sum_s\frac{b_s}{\sin^2\pi(z+s)},$$

and let Γ cross the negative axis at x^*, with heads $x_{s,n}=-s-n>x^*$.

Lemma 6.5 (Gram formula)**.** *Let $\kappa_{t,t'}(z)=1/((z-t)(z-t'))$ for nonnegative integers t,t'. Then*

$$\mathsf{L}_F[t,t']=\frac{1}{2\pi i}\int_\Gamma \kappa_{t,t'}E\,dz-\sum_{s,n}\bigl(b_s\kappa'_{t,t'}(x_{s,n})-a_s\kappa_{t,t'}(x_{s,n})\bigr).$$

Proof. Put $\tilde F(z) = \sum_s (a_s\psi(1-s-z) - b_s\psi'(1-s-z))$. The reflection formulas $\psi(1-w) = \psi(w) + \pi\cot\pi w$ and $\psi'(1-w) = \pi^2/\sin^2\pi w - \psi'(w)$ give $F = \tilde F + E$. The residues of $F\kappa_{t,t'}$ at t and t' give the Löwner entry, also for $t = t'$. At a head, ψ has residue -1 and ψ' has principal part $(z-x)^{-2}$, which gives the atom terms. The function $\tilde F$ is holomorphic to the left of Γ and $O(1/|z|)$ there by the balance, so its integral against $\kappa_{t,t'}$ vanishes. The function E decays like $e^{-2\pi|\operatorname{Im} z|}$, again by the balance. □

The atom term at a double head is a symmetric bilinear form of rank two in the row and column functions, and it splits into two rank-one terms. For functions g and h put $\hat g = g' - \frac{a_s}{2b_s}g$ and define $\hat h$ likewise. Then

$$-b_s(gh)' + a_s gh = -\frac{b_s}{2}[(g+\hat g)(h+\hat h) - (g-\hat g)(h-\hat h)]. \tag{6.3}$$

So the measure Ω of (6.1) is replaced by a real measure on a parameter space with two labels at each double head, carrying the weights $-b_s/2$ and $b_s/2$, and one label at each simple head. Andréief's identity holds on any measure space, so it writes every minor of L_F as an integral of products of two determinants. A repeated label gives two equal rows, so its term vanishes.

The determinants now contain derivatives of Cauchy kernels at the double heads. We compare them with a frame that has only simple poles. A *reference frame* for F is a frame $\mathcal{R}$ with simple poles that contains the shifts of F, together with one further shift s^* for each double shift s, such that the head of s^* lies within distance $\frac12$ of that of s, on the same side of x^*. For $F^{(2)}_{v,u}$ with $u \neq 0$ we take the shifts 1, $\frac34$, $\frac12$, with heads $-n$, $-n+\frac14$, $\frac12 - n$. For $F^{(2)}_{1,0}$ we take the shifts 1, $\frac12$ of the function $F_{1/2}$.

Lemma 6.6 (jets)**.** *Let z be an injective list of k labels, let the rows be k distinct nonnegative integers t_i, and let x^0 be the list of the heads of $\mathcal{R}$ matched to z. Then the determinant of the functions of (6.3) at z is at most $e^{O(k\log(r+k))}$ times the Cauchy determinant $|\det[(x^0_l - t_i)^{-1}]|$, uniformly for $k = O(r)$.*

Proof. A row at a double head is $\kappa\,\upsilon_t + \upsilon_t'$ evaluated at the head, with a constant κ. So the determinant is a product of operators $\kappa_l \pm \partial_l$ applied to the Cauchy determinant $\det[(x_l - t_i)^{-1}]$ in independent variables x_l. Cauchy's estimate on circles of radius $\frac18$, one variable at a time, bounds it by $\prod_l(|\kappa_l|+8)$ times the supremum of the Cauchy determinant over the polydisc. On the polydisc compare each factor of Cauchy's product formula with its value at x^0. Two variables on one double head move by at most $\frac38$ each and stay at distance at most $\frac14$, which is their distance at x^0. Every other ratio is at most $(1 + e_a/|x^0_a - x^0_c|)(1 + e_c/|x^0_a - x^0_c|)$ with $e_a \le \frac38$. The heads lie on the lattice $\frac14\mathbb{Z}$ with a bounded number per unit, so the sum of the reciprocal distances is $O(\log r)$ for each variable. The contour points lie at distance

bounded below from the heads, by the choice of Γ. Taking logarithms gives the bound. □

Theorem 6.7 (decay for frames)**.** *Let F be a frame and $\mathcal{R}$ a reference frame for it, or F itself if all orders are 1. Suppose that $\mathcal{R}$ has a certificate for Γ and ν_c as in Theorem 6.2, with $|\mathcal{T}|$ replaced by d, the number of poles of F per period counted with order. Then the conclusion of Theorem 6.2 holds for every member of the alternating geometry of parameter c for F, with the same D and ρ.*

Proof. By Lemma 6.5, (6.3) and Andréief's identity, each minor is an integral over the labels of products of two determinants. Lemma 6.6 bounds each of them by the Cauchy expression of $\mathcal{R}$ at the matched heads, at a cost $e^{O(r\log r)}$. The atom weights are bounded and $|E| \le C_E e^{-2\pi|\operatorname{Im} z|}$. These are the only inputs of the energy argument above. Each double head carries two labels, so the atom term is weighted by d. □

The certificate of the second plane is that of the reference frame $(1, \frac{3}{4}, \frac{1}{2})$ with $d = 3$. The certificate for $\zeta(3)$ is that of the function $F_{1/2}$ with $d = 2$. It is the certificate of the row for π^2, unchanged.

7. The first plane

In this section we prove Theorem 1.1 and parts (iii)–(v) of Theorem 1.2. We use the frames $F^{\pm}_{a,b,c}$ of §3.2 with the alternating geometry of parameter $c = \frac{1}{6}$, so $\alpha = \frac{1}{12}$ and $m = \lceil r/12 - \frac{1}{2} \rceil$. The point of the section is that the denominator exponent and the decay certificate do not depend on the weights (a, b, c). Nonvanishing is Theorem 5.2 with $d = 3$, so one deleted node suffices.

7.1. **Polar data.** Fix a frame $F^{\pm}_{a,b,c}$ with integer weights. Let $\varrho = 2$ on S^+ and $\varrho = 1$ on S^-. For $j \ge 1$ the increment is

$$\delta(j-1) = \frac{3a}{3j-\varrho} + \frac{2b}{2j-1} + \frac{c}{j},$$

and $-\delta'(j-1) = \frac{9a}{(3j-\varrho)^2} + \frac{4b}{(2j-1)^2} + \frac{c}{j^2}$. Let $p \ge 5$ be a prime with $3\bar{N} < p^2$. Call an integer t *singular* at p if p divides $t(3t-\varrho)(2t-1)$. The singular integers form the three lattices $p\mathbb{Z}$, $k_A + p\mathbb{Z}$ and $k_B + p\mathbb{Z}$, where $k_A \in [1, p)$ solves $3k_A \equiv \varrho \pmod p$ and $k_B = \frac{p+1}{2}$. We write $k_p = \min(k_A, k_B)$ for the first singular index. Then $k_A/p \to \kappa_A$ with

$$\kappa_A = \tfrac{1}{3} \text{ on } S^+,\ p \equiv 1\ (6) \text{ and on } S^-,\ p \equiv 5\ (6), \qquad \kappa_A = \tfrac{2}{3} \text{ otherwise,}$$

and $\kappa_B = \frac{1}{2}$. Passing from S^+ to S^- exchanges the two classes of primes modulo 6.

Lemma 7.1 (polar data)**.** *For every frame on $S^{\pm}$ with integer weights and every $1 \le j \le \bar{N}$, the increment above has p-adic valuation at least -1, and its derivative at least -2. If j is not singular, both are p-integral.*

Proof. Each of j, $3j-\varrho$ and $2j-1$ is smaller than $3\bar{N}<p^2$. So its valuation is at most 1, and it is 0 if j is not singular. The weights are integers. □

The lemma uses nothing about the weights but their integrality. If p divides a weight, the corresponding pole may disappear, and the lemma still holds since it gives only lower bounds. Rational weights are cleared first. Multiplying a frame by a nonzero integer multiplies ξ by that integer, which does not change μ.

7.2. **Local bounds for every frame.** For $x\neq y$ put

$$w(x,y)=\mathbf{1}[\text{a singular index lies in }(\min(x,y),\max(x,y)]]+\mathbf{1}[p\mid x-y],$$

and put $w(t,t)=2\cdot\mathbf{1}[t\geq k_p]$. By Lemma 7.1, the entry of $\Phi_{r,I}$ at (x,y) has Gauss valuation at least $-w(x,y)$. Lemma 4.1 then applies with $W=w$. We use four facts about w: $w\leq 2$, $w(x,y)\leq 1$ if $x\neq y$ and $p\nmid x-y$, $w(x,y)=0$ if $x\neq y$ are both below k_p, and $w(t,t)=0$ if $t<k_p$.

Proposition 7.2. *Let $p\geq 5$ be a prime with $3\bar{N}<p^2$. For every frame on $S^{\pm}$ and every member, $e_p(r,I)$ is at most each of the following, under its condition.*

	bound	*condition*
(T1)	$2r+2m$	*none*
(T1′)	$2r+2m-\|m\|_p$	*none*
(D2)	$2r-2(k_p+p-r)_+ +2m+(\min(\bar{N},p+k_p)-r)_+$	$2m<p\leq r$
(D4)	$2(r-\min(k_p,r))+2m+(\bar{N}-p)_+$	$p>r$
(D6)	$(\bar{N}-k_p)_+$	$p\geq\bar{N}$, $k_p\geq r$
(W)	$2r+2m-(t_2-t_1)$ *for each window of Theorem 4.7*	*as there*

In (W) the cuts are the three lattices of singular indices and the residue of r.

Proof. (T1) and (T1′) are proved as in §4.4.1. For the others we give potentials, with one function π used on rows and on columns. For (D2), $\pi=0$ on the common nodes $t<k_p$ with $t+p\geq r$, $\pi=1$ on the other common nodes, and $\pi(t)=1+\mathbf{1}[t-p<k_p]$ on the border. For (D4), $\pi(t)=\mathbf{1}[t\geq k_p]$ on the common nodes and $\pi(t)=1+\mathbf{1}[t\geq p]$ on the border. For (D6), $\pi=0$ on the common nodes and $\pi(t)=\mathbf{1}[t\geq k_p]$ on the border. In each case the four facts give $w\leq\pi\oplus\pi$ entrywise, and the sums are the stated values. For (W), the proof of Theorem 4.7 uses the function only through Lemma 7.1 and the cuts. Between two consecutive cuts the step function V and the diagonal residues are constant, which is what the translation of kernel vectors needs. No relation among the weights is used. □

None of these bounds reads the weights. They depend on (r, m, p) and on the class of p modulo 6 only.

7.3. **The denominator exponent.** Homogenise as in §4.5, with $k_A \mapsto \kappa_A\theta$, $k_B \mapsto \kappa_B\theta$ and $\theta = p/r$. Let φ_1 and φ_2 be the minima of the homogenised bounds of Proposition 7.2 for $\kappa_A = \frac{1}{3}$ and $\kappa_A = \frac{2}{3}$, with windows for $n \le 16$. Put $t_0 = \alpha/33$ and

$$\Lambda^* = (2 + 2\alpha)\, t_0 + \frac{1}{2}\int_{t_0}^{9} (\varphi_1 + \varphi_2)\, d\theta.$$

On $(t_0, \alpha/17]$ only (T1′) and the windows with $n = \lfloor m/p \rfloor$ act. On each strip $(\frac{\alpha}{n+1}, \frac{\alpha}{n}]$ the window lowers the sawtooth by a triangle of area $\alpha^2/(2n(n+1)(2n+1))$. So Λ^* is a rational number, and

$$\Lambda^* = 3.22274324065\ldots$$

Primes of the classes 1 and 5 modulo 6 each have density $\frac{1}{2}$, and S^- exchanges them, so Λ^* serves both supports. Above $\theta = \frac{7}{2}$ both profiles vanish.

Theorem 7.3 (denominators on the first plane)**.** *For every frame on $S^{\pm}$, every $\epsilon > 0$ and every even $r \ge R(\epsilon)$, every member satisfies* $\log \operatorname{den} P_{r,I} \le (\Lambda^* + \epsilon) r^2$. *The threshold $R(\epsilon)$ does not depend on the frame.*

Proof. Take the product over primes of the bounds of Proposition 7.2. For the primes with $p^2 \le 3\bar{N}$, and for 2 and 3, every entry has valuation at least $-2\log(3\bar{N})/\log p$. These primes contribute $O(r^{3/2})$. Each bound is built from affine functions of r, m, p, k_A, k_B and $p\lfloor r/p \rfloor$ by max, min and truncation, with bounded coefficients. So it differs from r times its homogenisation by a bounded amount. Each integer condition differs from its homogenised form only for p in an interval of bounded length, where (T1) costs $O(r \log r)$. The prime number theorem in progressions modulo 6 then gives the claim, as in Theorem 4.14. Primes $p \le t_0 r$ pay (T1), which gives $(2 + 2\alpha) t_0 r^2$. □

On $\theta \le \alpha$ the profile has a closed form. Only (T1), (T1′) and the windows act there, and with $s = 1/\theta$ one has $\varphi(\theta) = 2 + 2\alpha - \theta\, \Gamma_\kappa(s)$, where $\Gamma_\kappa(s)$ is the larger of $\|\alpha s\|$ and the longest gap of the lattice $\{0, \kappa_A, \kappa_B\} + \mathbb{Z}$ inside a window determined by the fractional parts of s and αs. This function has period 12 in s. The formal proof uses this to check one period of 81 cells per class, instead of the pieces of every strip.

7.4. **Decay.** Let Γ be the polygon of the formal proof for the alternating geometry of parameter $\frac{1}{6}$, crossing the real axis at distance $\frac{1}{4}$ from the poles of both supports.

Proposition 7.4. *There is a trial measure with* 25 *polynomial pieces on Γ, with a cover by cells satisfying the checks of §6.4 with $\varepsilon_0 = 10^{-6}$ and $|\mathcal{T}| = 3$,*

for which Theorem 6.2 gives

$$D = \frac{99853629795114224867}{28125000000000000000} = 3.55035\ldots,$$
$$\rho = \frac{33845462710966783}{5\cdot 10^{15}} = 6.76909\ldots$$

The certificate contains no weight. By §6.6 the decay estimate therefore holds for every frame on S^+ and on S^- with these D and ρ. The weights change only the threshold in r.

7.5. **Proof of Theorem 1.1.** Let a and b be rationals, not both zero. If $a = 0$, the number is a nonzero rational multiple of π^2, and Theorem 1.4, proved in Section 10, gives $\mu \le 6.5253$. Let $a \ne 0$ and $t = b/a$. By Lemma 3.2 there is a frame on S^+ or S^- whose ξ is a nonzero rational multiple of $L + t\pi^2$. Theorem 7.3 gives the denominators with Λ^*, Proposition 7.4 the decay, and Theorem 5.2 nonvanishing with one deleted node. The deduction of Theorem 10.1 at the even scales gives

$$\mu(\xi) \le \frac{\rho}{D - \Lambda^*} = 20.66216\,8\ldots < 20.6622.$$

The strict inequality gives the bound $|aL + b\pi^2 - p/q| > q^{-20.6622}$ for all large q, and in particular the irrationality. □

The thresholds depend on (a, b) through the height of the weights. No uniformity in (a, b) is claimed.

7.6. **Level returns.** At special weights the partial sums of the increment return to an earlier level, and then more entries are p-integral than Proposition 7.2 uses. Fix a frame on $S^{\pm}$, a class of primes modulo 6, and a prime $p \ge 7$ with $3\bar{N} < p^2$ and $2m < p$. Every singular index $t \le \bar{N}$ has $3t - \varrho$, $2t - 1$ or t equal to p times an integer prime to p and smaller than p. Hence

$$f_t = \frac{V(t)}{p} + (\text{a } p\text{-integer}), \qquad M[t,t] = \frac{U(t)}{p^2} + (\text{a } p\text{-integer}),$$

where the *level* $V(t)$ and $U(t)$ are rational numbers. They are sums of the terms $\frac{3a}{3l+n}$, $\frac{2b}{2l+1}$, $\frac{c}{l+1}$, respectively of their squared analogues with a sign, over the singular indices up to t. Here $n \in \{1, 2\}$ is fixed by the class. So V and U are constant between consecutive singular indices, and they depend on p only through the class. Along the first period V steps by the residues at the three poles. On S^+, for $p \equiv 1 \pmod 6$, the poles come in the order k_A, k_B, p with residues $3a$, $2b$, c, so the level after two poles is $3a+2b$, which is 0 exactly on the line of L. For $p \equiv 5 \pmod 6$ the order is k_B, k_A, p with residues $2b$, $\frac{3}{2}a$, c. A frame for which a partial sum of these residues vanishes has a *level return*. The frames $(2, -1, -1)$, $(1, -2, 1)$ and $(4, -3, -1)$ have level returns, with $b + \frac{a}{2} = 0$, $2a + b = 0$ and $2b + \frac{3}{2}a = 0$.

Lemma 7.5 (period sums)**.** *Let $t' = t + sp$ with $1 \le s < p$. Then $f_{t'} - f_t = (V(t') - V(t))/p + \Delta$ with $v_p(\Delta) \ge 1$.*

Proof. The polar terms give $(V(t') - V(t))/p$ exactly. For each shift, every block of p consecutive indices runs once through the residues modulo p. The nonpolar terms of a block therefore contribute a multiple of $\sum_{v \in \mathbb{F}_p^\times} v^{-1} \equiv 0 \pmod{p}$. □

So two congruent nodes at the same level have a p-integral entry, not one of valuation -1. A second saving comes from row operations inside a residue class. Let x be a border node and $a < x$ a common node with $x - a = jp$. Suppose that $V(x) \neq V(a)$, $U(a) \neq 0$, and that

$$\lambda_x = \frac{V(x) - V(a)}{j\,U(a)}$$

satisfies $\lambda_x \neq 1$ and has no prime factor larger than 5 in its denominator. Then λ_x is p-integral. We call (x, a) a *pivot pair*, and we put $V^*(x) = (V(x) - \lambda_x V(a))/(1 - \lambda_x)$. Pivot pairs whose border node is a column are defined in the same way. Each border node lies in at most one pivot pair, and $V^* = V$ on the nodes in no pair. Define weights w' on rows times columns as follows.

(i) For $x \not\equiv y \pmod{p}$, $w'(x, y) = \mathbf{1}[V^*(x) \neq V^*(y)]$.
(ii) For a pivot pair, the entry at the border node and its common node has $w' = 0$.
(iii) For other $x \equiv y$, $w'(x, y)$ is the largest $w_0(x_1, y_1)$ with x_1 equal to x or its common node and y_1 equal to y or its common node. Here $w_0(x_1, y_1) = 2 \cdot \mathbf{1}[V(x_1) \neq V(y_1)]$ for $x_1 \neq y_1$, and $w_0(t, t) = 2 \cdot \mathbf{1}[U(t) \neq 0]$.

Lemma 7.6 (levels)**.** *If nonnegative numbers u on the rows and u' on the columns satisfy $w' \leq u + u'$ entrywise, then $e_p(r, I) \leq \sum u + \sum u' + 2|I|$ for every member.*

Proof. Subtract λ_x times the row of a from the row of x for every pivot pair, and likewise for columns. The pivots are common nodes, which are never modified themselves, so these operations are unimodular and the determinant is unchanged as a polynomial in the variable. It suffices to bound the Gauss valuation of each new entry from below by $-w'$. For congruent $x \neq y$ without pivots, Lemma 7.5 gives $M[x, y] = (V(y) - V(x))/(sp^2) + \Delta/(sp)$, so the valuation is at least $-2 \cdot \mathbf{1}[V(x) \neq V(y)]$. With pivots the entry is a combination of at most four such entries with p-integral coefficients, which gives (iii). For a pivot pair the entry is $M[x, a] - \lambda_x(M[a, a] + x_{\mathrm{var}})$, where x_{var} is the variable. The terms of order p^{-2} cancel by the choice of λ_x, the rest is p-integral, and the variable has the p-integral coefficient $-\lambda_x$. This is (ii). For $y \not\equiv x$ write $\frac{1}{y-x} = \frac{1}{y-a} + \frac{jp}{(y-a)(y-x)}$. Then

$$M[x, y] - \lambda_x M[a, y] = \frac{(1 - \lambda_x)(V(y) - V^*(x))}{p\,(y - a)} + (\text{a } p\text{-integer}),$$

since $y - a$ and $y - x$ are units. If y is pivoted too, the same identity applied to its column gives the leading term $(1 - \lambda_x)(1 - \lambda_y)(V^*(y) - V^*(x))/(p(y - a))$.

This is (i). Finally, a deleted common node may be the common node of at most two pivot pairs, since $2m < p$. Those pairs are undone, which raises the bound by at most 2 per deleted node. □

The bound of Lemma 7.6 with no pivot pairs is already better than Lemma 4.1 at a level return. The pivots add more. They are the assignment form of the rank of the reduction modulo p on the fibres where the level returns. For instance, take the frame $(2,-1,-1)$ on S^+ and $p \equiv 5 \pmod 6$. The poles fall at about $\frac{p}{2}$, $\frac{2p}{3}$, p, $\frac{3p}{2}$, with residues -2, 3, -1, so V returns to 0 at p. A border node in $[\frac{3p}{2}, \frac{5p}{3})$ has level $-\frac{2}{3}$. Its common node in $[\frac{p}{2}, \frac{2p}{3})$ has level -2 and $U = 4$. Then $\lambda = \frac{1}{3}$ and $V^* = 0$, so after the row operation the border node behaves like a node at the returned level.

Theorem 7.7 (the special points)**.** *For the frames* $(2,-1,-1)$, $(1,-2,1)$ *and* $(4,-3,-1)$*, on* S^+ *and on* S^-*, the minimum of the bounds of Proposition 7.2 and Lemma 7.6 gives the denominator exponents*

$$\Lambda = 3.06684046\ldots, \qquad 3.10960666\ldots, \qquad 3.17760435\ldots$$

Proof. On each cell of the common refinement of the conditions, the nodes split into boundedly many classes by residue interval, parity and period. Pivot pairs, λ and V^* are constant on these classes. Potentials constant on the classes are affine in $\theta = p/r$, with a bounded rounding error per prime, as in Theorem 7.3. The potentials of the formal proof take half-integer values. They are verified on every pair of classes, and the minimum with the plane bounds is integrated as there. All pivots used have $\lambda \in \mathbb{Z}[\frac{1}{30}]$, so every prime $p \geq 7$ is admissible. The level branches act only for $\theta \geq \frac{3}{10}$. The frames on S^- give the same profiles with the classes exchanged. □

Proof of Theorem 1.2(iii)–(v). The frame $(2,-1,-1)$ has $\tau = \frac{2}{27}$. On S^+ its number is $9(L + \frac{2\pi^2}{27}) = \frac{1}{2}\Psi(\frac{1}{6}, \frac{1}{3})$, and on S^- it is $-9(L - \frac{2\pi^2}{27}) = -\frac{1}{2}\Psi(\frac{2}{3}, \frac{5}{6})$. The frame $(1,-2,1)$ has $\tau = -\frac{1}{27}$ and gives $\Psi(\frac{1}{2}, \frac{2}{3})$ on S^+ and $\Psi(\frac{1}{3}, \frac{1}{2})$ on S^-, up to sign. The frame $(4,-3,-1)$ has $\tau = \frac{1}{18}$ and gives $L \pm \frac{\pi^2}{18}$. With the denominator exponents of Theorem 7.7, the decay of Proposition 7.4 and Theorem 5.2, the deduction gives

$$\frac{\rho}{D - \Lambda} = 13.99987\,8\ldots, \qquad 15.35831\,0\ldots, \qquad 18.16002\,2\ldots,$$

below 14.000, 15.359 and 18.161. □

The point $t = 0$ is the frame $(2,-3,1)$, which has a level return after two poles in the class $p \equiv 1 \pmod 6$. It is treated with its own geometry in Section 8. The points $t = \pm\frac{1}{9}$ are given by the differences $F_{1/3}$ and $F_{2/3}$ of two digamma functions, with their own geometry, in Section 10.

8. The point L of the first plane

For $\Psi(\frac{1}{3}, \frac{2}{3})$ we use the function F_χ of (2.1), which combines three digamma terms, with the alternating geometry of parameter $c = \frac{1}{2}$ at all

even scales r. Then $\alpha = \frac{1}{4}$ and $m = \lceil r/4 - \frac{1}{2} \rceil$. The decay and nonvanishing estimates of Theorems 6.2 and 5.1 apply unchanged. The denominators need more care. At a prime p the increments $g(k)$ of §3 have poles in three residue classes instead of two, so the local bounds depend on two first poles. This section describes the two poles, the bounds we use, and the resulting constant.

8.1. **The first poles.** Let $p > 3$ be a prime. The increment $g(k) = \frac{1}{3k-2} - \frac{1}{2k-1} + \frac{1}{6k}$ has a pole modulo p exactly when p divides k, $3k-2$ or $2k-1$. Let k_A and k_B be the least $k \geq 1$ with $p \mid 3k-2$ and with $p \mid 2k-1$. Then

$$k_B = \frac{p+1}{2}, \qquad k_A = \begin{cases} \frac{p+2}{3}, & p \equiv 1 \pmod 6, \\ \frac{2p+2}{3}, & p \equiv 5 \pmod 6. \end{cases}$$

The residues of $p\,g(k)$ at k_A and k_B are 1 and -1 if $p \equiv 1 \pmod 6$, and $\frac{1}{2}$ and -1 if $p \equiv 5 \pmod 6$. In the first case $k_A < k_B$ and the two residues cancel. So the partial sums $\sum_{k \leq t} g(k)$ are p-integral for $t < k_A$, and again for $k_B \leq t < p$.

Both first poles are affine in one index. Define k_p by $6k_p = jp + 5$ with $0 < j < 6$. Then $j = 1$ for $p \equiv 1 \pmod 6$ and $j = 5$ for $p \equiv 5 \pmod 6$, $k_p = \frac{j}{6}p + O(1)$, and

$$(k_A, k_B) = \begin{cases} (2k_p - 1,\ 3k_p - 2), & p \equiv 1 \pmod 6, \\ (2k_p - p - 1,\ 3k_p - 2p - 2), & p \equiv 5 \pmod 6. \end{cases}$$

Thus j plays the role of the root class of §4, with q replaced by 6.

8.2. **The local bounds.** Fix a prime $p > 6$ with $6\bar{N} < p^2$. Every pole below $\bar{N}$ is then simple. Call a node t *regular* if $t < \min(k_A, k_B)$, or if $p \equiv 1 \pmod 6$ and t is a border node with $k_B \leq t < p$. At a regular node the partial sum of the increments up to t is p-integral. When t is a common node, its diagonal entry is p-integral too.

Proposition 8.1. *Every member of the alternating geometry of parameter $\frac{1}{2}$ for F_χ satisfies $e_p(r, I) \leq \tau$ for each of the following τ.*

(i) The bounds (T1) and (T1′), namely $2(r+m)$ and $2r + 2m - \|m\|_p$.

(ii) If $\bar{N} \leq 2p$, the bound $2r + 2m - n_{\mathrm{reg}}$. Here n_{reg} is the number of pairs formed by a regular node t and a side on which t lies, row or column, such that $t + p$ is not a node of the other side.

(iii) If $\bar{N} < \min(k_A, k_B)$, the bound 0.

(iv) The bound $2r + 2m - (t_2 - t_1)$ of Theorem 4.7, for every n and every window $[t_1, t_2)$ as there, with the cuts $t \equiv 0$, $t \equiv k_A$, $t \equiv k_B$ and $t \equiv r \pmod p$.

Proof. The bounds (i) are proved as in §4.4.1. For (ii) we apply Lemma 4.1 with the following potentials. At a node that is not regular, the potential is 1 on its row and on its column. At a regular node t, the potential on its row

is 1 if $t+p$ is a column node, and 0 otherwise. The potential on its column is 1 if $t+p$ is a row node, and 0 otherwise. Their sum is $2(r+m)-n_{\mathrm{reg}}$, and they dominate the weights W. Indeed, a diagonal weight is 0 at a regular common node and at most 2 in general. Two nodes that are not congruent modulo p have weight at most 1, since all poles are simple. Their weight is 0 if both are regular. Two distinct congruent nodes have weight at most 2. Since $\bar{N} \le 2p$ and regular nodes are smaller than p, at most one of them is regular. Suppose that the row node t of such an entry is regular. Then its column node is $t+p$, so the potential on the row of t is 1. The case of a regular column node is symmetric.

For (iii), the condition $\bar{N} < \min(k_A, k_B)$ implies $\bar{N} < p$. Then no index below $\bar{N}$ is a pole and no two nodes are congruent modulo p, so every entry is p-integral. For (iv), the proof of Theorem 4.7 uses F only through its poles. The step function V and the diagonal residues are constant between consecutive cuts. This is what the translation between the fibres of the window needs. □

The bounds (T2)–(T5), (LP1), (LP2), (R) and (S) of Theorem 4.5 are not used for F_χ.

8.3. **The denominators.** The tables for Λ of §4.6 and Theorem 4.14 apply to F_χ with q replaced by 6. The root class is $j \in \{1,5\}$ as above, and the bounds are those of Proposition 8.1. These bounds are expressions of the class $\mathcal{E}$ in r, p, m, k_A, k_B. On each class of primes modulo 6 the substitution above makes them expressions in r, p, m, k_p. Their homogenisations are taken with $\kappa = j/6$. The proof of Theorem 4.14 uses F only through these bounds and the root class. It goes through with the bound (iii) in place of (T5) beyond the end of the table.

The table of the formal proof starts at $\theta_0 = \frac{1}{68} = \alpha/17$. Below it the trivial bound $\ell_0 r$ with $\ell_0 = \frac{5}{2} = 2 + 2\alpha$ is used. On $(\alpha/17, \alpha]$ the table uses the bounds (i) and the windows (iv) with $1 \le n \le 16$. Beyond α it uses (i), (ii), the windows (iv) with $n = 0$ between consecutive cuts, and the bound (iii), which ends the table. The table is admissible for the two classes $j = 1, 5$. The arithmetic estimate therefore holds for every even r and every member with $|I| \le 64$, with the denominator exponent

$$\Lambda_\chi = 4.09592776731\ldots$$

This is an explicit rational number, whose numerator and denominator have 117 and 116 digits.

8.4. **Decay and nonvanishing.** The decay estimate is Theorem 6.2 for the function F_χ, with $|\mathcal{T}| = 3$ and $c = \frac{1}{2}$. The node measure $\nu_{1/2}$ has mass $\frac{5}{4}$ and energy

$$\mathcal{I}(\nu_{1/2}) = \frac{75}{32} - \frac{9}{8}\log\frac{3}{2} - \frac{1}{16}\log 2.$$

A trial measure on a polygon, with a cover by cells satisfying the checks of §6.4, gives

$$D = \frac{4643511791526209475773}{10^{21}} = 4.64351\ldots,$$
$$\rho = \frac{37521870783566737}{5\cdot 10^{15}} = 7.50437\ldots$$

Nonvanishing is Theorem 5.1 for F_χ. Its increment Δ_F has its pole at -1 isolated in its class modulo $\mathbb{Z}$, and the depth is $K \le 64$. The deduction of Section 10, at the even scales, gives

$$\mu\big(\Psi(\tfrac13,\tfrac23)\big) \le \frac{\rho}{D-\Lambda_\chi} = 13.704516\ldots < 13.7046,$$

since $\Psi(\frac13,\frac23) = 36\,\xi_\chi$.

9. The second plane

In this section we prove Theorem 1.3 and the bound for $\zeta(3)$ in Theorem 1.4. We use the frames $F^{(2)}_{v,u}$ of §3.2 with integers $v \neq 0$ and u, and we put $\varepsilon = \mathbf{1}[u \neq 0]$. For $u \neq 0$ we take the alternating geometry of parameter $c = \frac{6}{41}$, so $\alpha = \frac{3}{41}$. For $u = 0$ we take $c = \frac{3}{10}$. In both cases $2m \le r$. The new feature is the double pole at the shift 1. The entries of the Löwner matrix then have valuation down to -3, and most of the work goes into recovering what is lost against a simple pole.

9.1. **Cells and weights.** Let $p \ge 5$ be a prime with $p^2 > 2\bar N + 2p$. The increment is

$$\delta(k-1) = \frac{u}{k} - \frac{2u}{2k-1} - \frac{v}{k^2} \qquad (k \ge 1).$$

Its poles below $\bar N$ are the double poles $k = jp$ and, if $u \neq 0$, the simple poles $k = jp + \frac{p+1}{2}$. Write an integer $t \ge 0$ as $t = jp + \omega$ with $0 \le \omega < p$. We call j the *period* of t, $h = \mathbf{1}[\omega \ge \frac{p+1}{2}]$ its *half*, and $c(t) = (j,h)$ its *cell*. A pole lies at the start of every cell. Let $S(c)$ be the sum of the polar terms of δ up to the start of the cell c, and $G(c)$ the same for $-\delta'$. Then

$$f_t = S(c(t)) + \rho_t, \qquad e_t = G(c(t)) + \gamma_t,$$

with ρ_t and γ_t in $\mathbb{Z}_{(p)}$, and p^2S and p^3G in $\mathbb{Z}_{(p)}$. Modulo p the numbers p^2S and p^3G depend only on the period, because the simple poles have lower order than the double ones.

Lemma 9.1 (weights). *Let $W(t,t') = \max(0, -v_p(M[t,t']))$. Then $W(t,t') \le 3$ if $t \equiv t' \pmod p$, and for $t \not\equiv t'$*

$$W(t,t') \le W_3(c(t),c(t')), \qquad W_3(c,c') = \begin{cases} 0 & c = c', \\ 1 & \text{same period, other half}, \\ 2 & \text{other period}. \end{cases}$$

On the diagonal $W(t,t)$ is 0 on the cell $(0,0)$, at most 2ε on $(0,1)$, and at most 3 elsewhere. This holds also when p divides v or u.

Proof. For $t \neq t'$ the difference $t'-t$ is a unit unless $t \equiv t'$, and then $v_p(t'-t) = 1$ since $p^2 > \bar{N}$. The numerator $f_{t'} - f_t$ has valuation at least -2, at least -1 if t and t' lie in one period, and at least 0 if they lie in one cell. The diagonal is $-e_t$. The weights u and v are integers, so a prime dividing them only raises valuations. □

The cases $p \mid v$ need no separate treatment below. Each row we prove either follows from Lemma 9.1 alone, or is at least $2(r+m)$, which is the trivial bound when every entry has valuation at least -2. That is the case if $p \mid v$.

Proposition 9.2 (assignment rows)**.** *Assume $2m \le r$. For every member,*

	bound	*condition*
(P1)	$3(r+m) - \|m\|_p$	*none*
(P4)	$6(r+m-p) + \varepsilon p$	$p \le r$
(P5)	$\varepsilon r + 3(\bar{N} - p)$	$r \le p \le \bar{N}$
(P6)	$\varepsilon(2r + 2m - p + 1)$	$\bar{N} \le p \le 2r$
(P7)	$\varepsilon(\bar{N} - \frac{p-1}{2})_+$	$p \ge 2r$

Proof. All follow from Lemmas 4.1 and 9.1. For (P1) take the potentials of (T1′) shifted by $\frac{1}{2}$: on the rows of a residue class $\frac{3}{2} - \frac{1}{2}\sigma$ and on its columns $\frac{3}{2} + \frac{1}{2}\sigma$, where $\sigma = \operatorname{sign}(|X' \cap \text{class}| - |Y' \cap \text{class}|)$. Congruent pairs get 3 and the others at least 2. For the other rows take one function φ on rows and columns, with $\varphi = 0$ on the cell $(0,0)$, $\varphi = \varepsilon$ on $(0,1)$ and $\varphi = 3$ on the periods $j \ge 1$. Pairs inside the period 0 have weight at most $\varphi \oplus \varphi$, and any pair with a node in a later period has weight at most 3. Counting the common nodes twice and the border nodes once gives the four bounds. The hypothesis $2m \le r$ places the border in the period 0 for (P5) and above $\frac{p}{2}$ for (P7). □

Proposition 9.2 prices a prime $p \le 2m$ at $3(r+m) - \|m\|_p$, and a prime $2m < p \le r$ at no better. The double pole costs one more unit per node than a simple pole. The next two subsections recover most of this unit on the primes below $r + 2m$.

9.2. **Elimination of complete periods.** Let $\mathcal{O}$ be the discrete valuation ring of §4, in a variable X, and put $x = p^{-3}X$, so that a member is a polynomial in x. Fix $\kappa \ge 1$ and $k' \ge 0$ with $(\kappa + 2k')p \le \bar{N} + p$, and put $\bar{N}' = \max(\bar{N}, (\kappa+2k')p)$. Then $p^2 > 2\bar{N}'$. Let $H = M + p^{-3}X\mathrm{Id}$ on $[0, \bar{N}')$, so that H restricts to the matrices of the members. Extend the border alternately, so $\hat{X} = \{t \ge r : t - r \text{ even}\}$ and $\hat{Y} = \{t \ge r : t - r \text{ odd}\}$. The *pivot* consists of the complete periods below κp and of $2k'$ further complete periods:

$$\mathcal{P}_r = [0, \kappa p) \cup (\hat{X} \cap [\kappa p, (\kappa+2k')p)), \qquad \mathcal{P}_c = [0, \kappa p) \cup (\hat{Y} \cap [\kappa p, (\kappa+2k')p)).$$

Lemma 9.3 (displacement and bordering)**.** *For all nodes s, t one has $(s-t)H(s,t) = f_s - f_t$, whatever the diagonal. Suppose $H[\mathcal{P}_r, \mathcal{P}_c]$ is invertible, and let Z be its inverse. Let* Sw *be the Schur complement of $H[\mathcal{P}_r, \mathcal{P}_c]$ in the matrix bordered by unit vectors, as in Lemma 4.9. Then*

(a) every minor of H that contains $\mathcal{P}_r \backslash A_0$ and $\mathcal{P}_c \backslash A_0'$ equals $\pm \det H[\mathcal{P}_r, \mathcal{P}_c]$ times a minor of Sw *with rows $B_0 \cup A_0'$ and columns $B_0' \cup A_0$, where B_0 and B_0' are the remaining rows and columns.*

(b) there are vectors $g_u \in \mathbb{Q}(X)^2$, one for each label u, such that $(v-u)\operatorname{Sw}(u,v) = g_u \wedge g_v$ for all labels $u \neq v$.

Proof. The first identity is the definition of the Löwner matrix. With $\mathsf{T} = \operatorname{diag}(t)$ it says that $\mathsf{T}H - H\mathsf{T} = f\mathbf{1}^{\mathsf{T}} - \mathbf{1}f^{\mathsf{T}}$ has rank two. The bordered matrix satisfies the same equation with generators $(1, f_t)$ on actual nodes and 0 on the border, since $(s-s)\cdot 1 = 0$. A Schur complement of a block preserves this displacement structure, with generators $g = G - VZG_{\mathcal{P}}$. Laplace expansion along the unit border gives (a). □

The generators are explicit. On a label z outside the pivot $g_z = (1 - H(z,\mathcal{P}_c)Z\mathbf{1},\ f_z - H(z,\mathcal{P}_c)Zf)$, and on a pivot label t it is $((Z\mathbf{1})_t, (Zf)_t)$ up to sign. We rescale them as $\tilde{g}_u = (p^{-3}g_{u,1}, p^{-1}g_{u,2})$ on pivot labels and $\tilde{g}_u = (g_{u,1}, p^2 g_{u,2})$ on the others, and the pivot rows and columns of Sw by p^{-3}. The rescaled matrix $\widetilde{\mathrm{Sw}}$ then has entries $p^{-2}(\tilde{g}_u \wedge \tilde{g}_v)/(v-u)$.

The reduction of the pivot is governed by matrices on period indices. For $h \in \{0,1\}$ put $s_h(j) = p^2 S(j,h)$, $g_h(j) = p^3 G(j,h)$, $\Gamma_h(j) = (1, s_h(j))$, and

$$B^h(l,j) = \frac{s_h(j) - s_h(l)}{j-l}\ (l \neq j), \qquad B^h(j,j) = X + g_h(j).$$

Let $J_X = \{\kappa, \kappa+2, \ldots, \kappa+2k'-2\}$, $J_Y = J_X + 1$, and $B_X^h = B^h[[0,\kappa) \cup J_X, [0,\kappa) \cup J_Y]$. Write $\bar{B}_X$ for its reduction modulo p, which does not depend on h. If $t \equiv t' \pmod p$ then $p^3 H(t,t') \equiv B^h(j_t, j_{t'}) \pmod{p^3}$. If $t \not\equiv t'$ then $p^3 H(t,t') \in p\mathcal{O}$. Hence $p^3 H[\mathcal{P}_r, \mathcal{P}_c]$ is congruent modulo p to a direct sum of copies of $\bar{B}_X$ and its transpose, one for each residue class.

Theorem 9.4 (nondegeneracy)**.** *Let $\kappa \geq 1$, $k' \geq 0$ and $p > \kappa + 2k'$. Then $\det \bar{B}_X \neq 0$ in $\mathbb{F}_p(X)$.*

Proof. For $k' = 0$ the determinant is monic in X. Let $k' \geq 1$. The variable occurs only on the diagonal of the block $[0,\kappa)$, so $\det \bar{B}_X$ is $\pm \det \bar{B}[J_X, J_Y]$ times a monic polynomial, and it suffices that $D = \det[(h_j - h_l)/(j-l)]_{l \in J_X, j \in J_Y}$ is nonzero in $\mathbb{F}_p$, where $h_t = \sum_{i \leq t} i^{-2}$. Suppose that $c \neq 0$ lies in its kernel. Let $P = \prod_{j \in J_Y}(z-j)$, $\mathsf{u} = \sum_j c_j P/(z-j)$ and $\mathsf{w} = \sum_j c_j h_j P/(z-j)$. These polynomials over $\mathbb{F}_p$ have degree at most $k'-1$, $\mathsf{u} \neq 0$, and $\mathsf{w}(t) = h_t \mathsf{u}(t)$ on the $2k'$ consecutive nodes $\kappa, \ldots, \kappa+2k'-1$. Since $h_{t+1} - h_t = (t+1)^{-2}$, the polynomial

$$\mathsf{E}(z) = (z+1)^2\big(\mathsf{w}(z+1)\mathsf{u}(z) - \mathsf{w}(z)\mathsf{u}(z+1)\big) - \mathsf{u}(z)\mathsf{u}(z+1)$$

vanishes at the $2k'-1$ nodes other than the last. The top terms cancel, so $\deg \mathsf{E} \le 2k'-2$ and $\mathsf{E}=0$. Divide u and w by their greatest common divisor. At $z=-1$ the identity gives $\mathsf{u}(-1)\mathsf{u}(0)=0$. If $\mathsf{u}(x+1)=0$ with $x\neq -1$ then $\mathsf{u}(x)=0$, and conversely, by coprimality. So every element of $\mathbb{F}_p$ is a root of u, whose degree is less than p. This is a contradiction. □

The argument is that of Theorem 5.2, in characteristic p. The double pole at -1 is isolated in its orbit, and the orbit is now finite, which replaces the infinitely many roots by p roots.

Proposition 9.5 (generators)**.** *Let $p > \kappa + 2k'$. Then all $\tilde g_u$ lie in $\mathcal{O}^2$. There are vectors $\mathcal{V}(c) \in \mathcal{O}^2$ on the cells of the periods outside the pivot, of the period $\kappa-1$ and of the period $\kappa+2k'-1$, such that every label u in these periods satisfies $\tilde g_u \equiv \epsilon_u \mathcal{V}(c(u)) \pmod{p^2}$ with a unit $\epsilon_u \equiv 1 \pmod p$, and $\mathcal{V}(j,0) \equiv \mathcal{V}(j,1) \pmod p$. Consequently, for two such labels $u \neq v$,*

$$v(\widetilde{\mathrm{Sw}}(u,v)) \ge -3 \text{ if } u \equiv v \pmod p,$$
$$v(\widetilde{\mathrm{Sw}}(u,v)) \ge -W_3(c(u),c(v)) \text{ otherwise,}$$

and $v(\widetilde{\mathrm{Sw}}(t,t)) \ge -3$ on deleted nodes.

Proof. Let $Q = p^3 H[\mathcal{P}_r, \mathcal{P}_c]$. By Theorem 9.4 and the direct sum above, Q is invertible over $\mathcal{O}$. We look for the solution Θ of $Q\Theta = (\mathbf{1}, p^2 f)$ modulo p^2. Modulo p^2 the right side is a function of the cell, and modulo p of the period. Put $\Theta(t) = \epsilon(t) Y^h(j_t)$ with $Y^h = (B^h_X)^{-1}\Gamma_h$ on a class oriented like $\hat X$, and the transposed matrix on the others. The congruent part of $Q(u,\cdot)\Theta$ is then $\epsilon B^h(j_u,\cdot)Y^h$, which is $\epsilon\Gamma_h(j_u)$ on the pivot. The non-congruent part lies in $p\mathcal{O}$, and modulo p^2 it equals

$$p \sum_{\omega' \neq \omega_u} \frac{1}{\omega' - \omega_u} \big(\Psi_{o(\omega')}[1,\cdot] - \bar s(j_u)\Psi_{o(\omega')}[0,\cdot]\big), \qquad \Psi_X = \bar\Gamma^{\mathsf T} \bar B_X^{-1} \bar\Gamma,$$

with $\Psi_Y = \Psi_X^{\mathsf T}$. Here we use that the pivot consists of complete periods, so every residue carries Ψ_X or Ψ_Y and nothing else. Since $\sum_{\omega'\neq\omega}(\omega'-\omega)^{-1} \equiv 0 \pmod p$, only the antisymmetric part $\Psi_X - \Psi_X^{\mathsf T}$ survives. It acts on 2-vectors as a scalar δ times the rotation, so the sum is $p\delta E(\omega_u)\bar\Gamma(j_u)$, where $E(\omega)$ is the sum of $(\omega'-\omega)^{-1}$ over the residues of one orientation. Taking $\epsilon = 1 - p\delta E$ gives $Q\Theta \equiv (\mathbf{1}, p^2 f) \pmod{p^2}$ on the pivot, and the same computation outside the pivot gives $\tilde g_z \equiv \epsilon(\Gamma(z) - B^h(j_z,\cdot)Y^h) \pmod{p^2}$. These are the vectors $\mathcal{V}$. They depend on the cell modulo p^2, and on the period modulo p, since B^h and Γ_h do not depend on h modulo p. For $k'=0$ we have $\delta = 0$.

By Lemma 9.3, $\widetilde{\mathrm{Sw}}(u,v) = p^{-2}(\tilde g_u \wedge \tilde g_v)/(v-u)$. The wedge lies in $p^2\mathcal{O}$ for two labels in one cell and in $p\mathcal{O}$ for two labels in one period. The difference $v-u$ has valuation 1 if $u \equiv v$ and 0 otherwise. The diagonal entry on a deleted node is $-p^{-3}Q^{-1}(t,t)$. □

9.3. **The rows of the second plane.** We now combine Lemma 9.3(a) with Proposition 9.5. A potential on the labels bounds the Leibniz terms of the minor of $\widetilde{\mathrm{Sw}}$, and the factor $\det H[\mathcal{P}_r, \mathcal{P}_c]$ has valuation exactly $-3|\mathcal{P}_r|$.

Theorem 9.6 (rows of the second plane)**.** *Let $p \geq 5$ with $p^2 > 2\bar{N} + 2p$ and $2m \leq r$. For every member the following hold.*

(i) (E_1) If $p > 2m$, then $e_p(r, I) \leq 3r + \sum u + \sum u'$ for all nonnegative potentials u on X' and u' on Y' with $W_3 \leq u \oplus u'$ on the border.

(ii) (E_2) If $p \leq r$ and $\bar{N} \leq 2p$, then (E_1) holds with $3r$ replaced by $3r - (2p - \bar{N})$.

(iii) (R) If $p \leq 2m$, write $2m = kp + s$ with $0 \leq s < p$, and let $E = [r, r+s)$ for even k and $E = [r+s, r+p)$ for odd k. For $t \in E$ let $c_b(t) \in \{0, 1, 2, 3\}$ be twice the period of t minus that of r, plus the half of t. Then, for each $w \in \{(2, 2, 0, 1), (2, 2, 1, 0), (0, 1, 2, 2), (1, 0, 2, 2)\}$,

$$e_p(r, I) \leq 3(r + m) - 3\|m\|_p + \sum_{t \in E} w(c_b(t)).$$

Proof. In each case $\kappa + 2k' \leq \bar{N}/p + 1 < p$, so Theorem 9.4 and Proposition 9.5 apply.

(i) Take $\kappa = \lfloor r/p \rfloor + 1$ and $k' = 0$. Since $2m < p$, no two border nodes are congruent. The nodes of $[\bar{N}, \kappa p)$ lie in one period with the border nodes beyond $\kappa p - p$. Give them the potentials $\frac{3}{2}$ on both sides, and the deleted nodes the potentials 3. Proposition 9.5 bounds every Leibniz term of the minor of $\widetilde{\mathrm{Sw}}$, and Lemma 9.3(a) with the count of the pivot gives the bound for the Gauss valuation. The coefficient of x^i carries an extra $3i \geq 0$.

(ii) Take $\kappa = 2$, $k' = 0$ and X replaced by pX. Then $\bar{B}_X = \left(\begin{smallmatrix} 0 & -1 \\ -1 & 2 \end{smallmatrix}\right)$, because there is no pole at $k = 0$. Its inverse has entry 0 at $(1, 1)$. So $Q^{-1}(t, t) \in p\mathcal{O}$ for t in the period 1, and the virtual nodes $[\bar{N}, 2p)$ need the potentials 1 only. This saves $2p - \bar{N}$.

(iii) Take $\kappa = \lfloor r/p \rfloor + 1$ and $k' = \lfloor k/2 \rfloor$. The labels outside the pivot are shifted down by $2k'p$. This keeps parity, residue and half, and it maps them onto the nodes of $[r, \bar{N} - 2k'p)$ together with the virtual nodes. The window $[r, \bar{N} - 2k'p)$ has length $p + s$ or s. Each residue class of the border has an even number of nodes except the $2\|m\|_p$ classes of E. The nodes outside E pair off at weight 3 per pair. On E we use the potential $w(c_b(t))$ on both sides. Each w in the list dominates W_3 on the four cells that E can meet. For the other nodes choose a period j_0 meeting E, such that $w \geq 1$ on the cells of E in the other period, if any. Give a node outside E the potential 1 if it lies in j_0 and 2 otherwise. Every congruent pair outside E has exactly one node in j_0, so its potentials sum to 3. All other pairs have $W \leq 2$, or $W \leq 1$ inside j_0, and the bound follows as in (i). □

For $k = 1$ part (iii) needs only the pivot of (i), with the weight 3 of congruent pairs kept. For $k \geq 2$ it needs the complete double periods. The pivot contracts the window of length $2m$ to one of length $p + s$ or s, and the cost of each eliminated pair of periods is exactly 3 per node.

9.4. **The denominator exponent.** Homogenise with $\theta = p/r$ as before. The profile $\mathcal{P}(\theta)$ is the minimum of the homogenised bounds of Proposition 9.2 and Theorem 9.6. For (E_1) and (E_2) the potentials are constant on the at most three cells cut from the border at the multiples of $\theta/2$. On the band $\frac{2\alpha}{k+1} < \theta \le \frac{2\alpha}{k}$, the bound (R) is

$$3 + \tfrac{3}{2}k\theta + \tau(E) \ (k \text{ even}), \qquad 3 + 6\alpha - \tfrac{3}{2}(k+1)\theta + \tau(E) \ (k \text{ odd}),$$

where $\tau(E)$ is the least of the four sums $\sum_c w(c)|c|/2$ over the cells c of $E = [1, 1 + 2\alpha - k\theta]$, respectively $[1 + 2\alpha - k\theta, 1 + \theta]$. Between the points $\theta = 2/j$ and $\theta = 2(1 + 2\alpha)/j$ the cell type of E is fixed. So $\mathcal{P}$ is piecewise affine on $(2\alpha/18, \infty)$, with finitely many pieces. We use (R) for $k \le 17$. Below $2\alpha/18$ the profile is the sawtooth $3 + 3\alpha - \theta\|\alpha/\theta\|$ of (P1).

Theorem 9.7 (denominators on the second plane)**.** *Let $\alpha = \frac{3}{41}$ and $u \neq 0$. For every $\epsilon > 0$ and every even $r \ge R(\epsilon)$, every member with $|I| \le 1$ satisfies* $\log \operatorname{den} P_{r,I} \le (\Lambda_2 + \epsilon)r^2$, *where*

$$\Lambda_2 = \int_0^\infty \mathcal{P}(\theta)\, d\theta \le \frac{869210883039}{250000000000} = 3.476843532 1\ldots$$

The threshold does not depend on (u, v).

Proof. For each prime with $p^2 > 2\bar{N} + 2p$ the rows give $e_p(r, I) \le r\,\mathcal{P}(p/r) + O(1)$. For the rows (E_1), (E_2) and (R) the cells of the integer window and of its homogenisation differ by a bounded number of cells of bounded length, which take the potential 2. The hypotheses $p > \kappa + 2k'$ of Proposition 9.5 hold for r large, since $p > 2\alpha r/18$ on the bands used. For the other primes every entry has valuation at least $-3\log(2\bar{N})/\log p$, which costs $O(r^{3/2})$ in total. The prime number theorem then gives the integral, as in Theorem 4.14. The integral of the sawtooth contributes a multiple of $\log 2$, and the rest is an exact rational integral. Both are bounded by the stated rational number. □

Without (R) the same integral is $3.47980\ldots$, and the exponent below would be about 389.

9.5. **Decay and proof of Theorem 1.3.** By Theorem 5.2 with $d = 3$, one deleted node suffices for nonvanishing. The decay is Theorem 6.7 with the reference frame of shifts 1, $\frac{3}{4}$, $\frac{1}{2}$.

Proposition 9.8. *For the reference frame of shifts 1, $\frac{3}{4}$, $\frac{1}{2}$ and $c = \frac{6}{41}$ there is a trial measure with 49 polynomial pieces and a cover by 508 cells satisfying the checks of §6.4 with $\varepsilon_0 = 10^{-6}$ and $d = 3$, for which Theorem 6.2 gives $D \ge 3.4982052$ and $\rho \le 7.1600631$.*

Proof of Theorem 1.3. Let a and b be rationals, not both zero. If $a = 0$ the number is a nonzero rational multiple of π^2, and Theorem 1.4 gives $\mu \le 6.5253$. If $b = 0$ it is a nonzero rational multiple of $\zeta(3)$, and Theorem 1.4 gives $\mu \le 6.0382$. Let $ab \neq 0$, and choose integers $v \neq 0$ and $u \neq 0$ with $u/v = 6b/a$. By §3.2, the number ξ of $F^{(2)}_{v,u}$ is a nonzero rational multiple of

$a\zeta(3)+b\pi^2$. Theorem 9.7, Proposition 9.8 and Theorem 5.2 give the three estimates, and Theorem 10.1 gives

$$\mu(\xi) \le \frac{\rho}{D-\Lambda_2} \le 335.1827 < 335.20. \qquad \square$$

9.6. **The value $\zeta(3)$.** For $u=0$ the frame is $\psi'(z+1)$, with $d=2$ and $\xi=-2\zeta(3)$. Everything above holds with the half map set to 0. The cells are then the periods, W_3 takes the values 0 and 2, and $\varepsilon=0$ in Proposition 9.2. The rows (P4) and (P5) become $6(r+m-p)$ and $3(\bar{N}-p)_+$, and (P6), (P7) vanish. In (R) the four cells become two, and the potentials are $(0,0,2,2)$ and $(2,2,0,0)$. With $c=\frac{3}{10}$ the integral of the profile is

$$\Lambda_{\zeta(3)} = \frac{78871534239346376957366127790384501}{27370345362695344229217529371840000} + \frac{9}{200}\log 2 = 2.91283\,35313\ldots$$

The reference frame is the function $F_{1/2}$, and its certificate is that of the row for π^2 in Table 2. Theorem 5.2 with $d=2$ gives nonvanishing with one deleted node.

Proof of Theorem 1.4 for $\zeta(3)$. The three estimates above and Theorem 10.1 give $\mu(\zeta(3)) \le \rho/(D-\Lambda_{\zeta(3)}) \le 6.038153 < 6.0382$. The bound for π^2 is proved in Section 10, with the rows of Theorem 1.5. $\square$

10. The deduction and the remaining proofs

The three estimates combine as follows.

Theorem 10.1 (deduction)**.** *Let F be a frame with a geometry. Let K be a depth such that, for every large r and every rational x_0, some member $P_{r,I}$ with $|I| \le K$ satisfies $P_{r,I}(x_0) \neq 0$, as in Theorems 5.1 and 5.2. Suppose that the members satisfy $\log \operatorname{den} P_{r,I} \le (\Lambda+o(1))r^2$ and the decay estimate of Theorem 6.2 with constants D and ρ, and that $\Lambda < D$. Then ξ is irrational and $\mu(\xi) \le \rho/(D-\Lambda)$.*

Proof. Let $\eta = D-\Lambda$, and let $\mu_0 > \mu_1 > \rho/\eta$. Suppose that $|\xi - a/b| < b^{-\mu_0}$ with b large. Let r be the least admissible scale with $r \ge \mu_1 \log b/\rho$. With $x_0 = a/b$ there is a member with $P_{r,I}(a/b) \neq 0$. By Lemma 3.5, $|\operatorname{den}(P_{r,I})\, P_{r,I}(a/b)| \ge b^{-r}$. Write $a/b = \xi + h$. Then $|h| < b^{-\mu_0} \le e^{-\rho r}$, because $\rho r \le \mu_1 \log b + O(1) < \mu_0 \log b$ for large b. By the two estimates,

$$|\operatorname{den}(P_{r,I})\, P_{r,I}(a/b)| \le e^{(\Lambda+o(1))r^2} e^{-(D-o(1))r^2} = e^{-(\eta-o(1))r^2}.$$

Comparing the two bounds gives $(\eta-o(1))r^2 \le r\log b$. But $\eta r \ge (\mu_1\eta/\rho)\log b$ with $\mu_1\eta/\rho > 1$. This is a contradiction for large b. If $\xi = a/b$ were rational, then for every large r some member would satisfy $b^{-r} \le |\operatorname{den}(P_{r,I})\, P_{r,I}(\xi)| \le e^{-(\eta-o(1))r^2}$, which is impossible. So ξ is irrational. $\square$

Proof of Theorem 1.5, Theorem 1.2(ii) and Theorem 1.4 for π^2. For each of these rows of Table 1 the function is F_s of (2.1), and the geometry parameter and the constants are those of Table 2. The denominator exponent is given by Theorem 4.14 with the table for Λ of the formal proof. The decay rates are given by Proposition 6.3, and nonvanishing by Theorem 5.1. Theorem 10.1, applied at the even scales, gives $\mu(\xi) \le \rho/(D-\Lambda)$, and in every row $\rho/(D-\Lambda)$ is below the stated bound. By (2.1), ξ is a nonzero rational multiple of $\Psi(s,1)$. Since the inequality is strict, the definition of μ gives $|\xi - a/b| > b^{-\mu_0}$ for all integers a and all sufficiently large b. □

Theorem 1.2(i) is proved in Section 8, parts (iii)–(v) and Theorem 1.1 in Section 7, and Theorem 1.3 and the bound for $\zeta(3)$ in Section 9. In each case the bound is attained in the same strong form.

row	c	Λ	D	ρ	ρ/η	μ
plane 1	$\frac{1}{6}$	3.2227433	3.5503512	6.7690926	20.66217	20.6622
$L \pm \frac{2\pi^2}{27}$	$\frac{1}{6}$	3.0668405	3.5503512	6.7690926	13.99988	14.000
$L \pm \frac{\pi^2}{27}$	$\frac{1}{6}$	3.1096067	3.5503512	6.7690926	15.35832	15.359
$L \pm \frac{\pi^2}{18}$	$\frac{1}{6}$	3.1776044	3.5503512	6.7690926	18.16003	18.161
$(\frac{1}{3},\frac{2}{3})$	$\frac{1}{2}$	4.0959278	4.6435117	7.5043742	13.70452	13.7046
$(\frac{k}{3},1)$	$\frac{3}{10}$	3.3104458	4.0291249	6.8272548	9.499725	9.4998
plane 2	$\frac{6}{41}$	3.4768436	3.4982052	7.1600631	335.1827	335.20
$\zeta(3)$	$\frac{3}{10}$	2.9128336	3.9785713	6.4350874	6.038153	6.0382
$(\frac{1}{2},1)$	$\frac{3}{10}$	2.9923824	3.9785713	6.4350874	6.525207	6.5253
$(\frac{k}{4},1)$	$\frac{1}{5}$	3.5149036	3.7447213	7.1904273	31.28751	31.2876
$(\frac{k}{5},1)$	$\frac{2}{9}$	3.5128488	3.8135797	7.1655663	23.82717	23.8272
$(\frac{k}{7},1)$	$\frac{1}{6}$	3.6391158643	3.6432250312	7.4218615374	1806.172	1806.1719

TABLE 2. The data of the proofs. A pair (s,t) denotes the function of (2.1) for $\Psi(s,t)$, with k prime to the denominator. For each row the table gives the geometry parameter c, the denominator exponent Λ, the decay rate D, the neighbourhood exponent ρ and the exponent. The constants are rational numbers, rounded outward to the digits shown: Λ and ρ up, D down. The column ρ/η, with $\eta = D-\Lambda$, is the exact quotient, rounded up. The three rows below plane 1 share its decay certificate. The row for $\zeta(3)$ shares the certificate of the row $(\frac{1}{2},1)$.

Remark 10.2 (relation to linear independence criteria). The deduction is a one-dimensional version of the familiar scheme behind the criteria of Nesterenko [30] and their use by Ball and Rivoal [2]. The nonzero integer $b^{|C\setminus I|}\operatorname{den}(P_{r,I})\,P_{r,I}(a/b)$ plays the role of a small linear form with integer

coefficients. There are two differences. The lower bound for the form comes from an arithmetic nonvanishing statement rather than from an analytic asymptotic. The size r of the determinant is chosen in terms of b.

10.1. **Notes on the rows.** For the rows $(\frac{k}{q}, 1)$ the function is F_s of (2.1) with $s = k/q$, $\omega = q^{-2}$ and $\xi_s = -q^{-2}\Psi(\frac{k}{q}, 1)$. The table for Λ, the trial measure and the cover are data of the formal proof. All shifts with the same denominator share one table for Λ, indexed by the root classes, and one certificate. Indeed the bounds depend on the shift only through the root class, and the decay depends on it only through the separation of the poles from the contour. For $q = 4$ the root classes $j \equiv kp^{-1} \pmod 4$ of $k = 1$ and $k = 3$ exchange the two classes of primes modulo 4. For $q = 3$ the table uses the primes in the classes modulo 6, and the root classes $j = 1, 2$ correspond to first poles near $p/3$ and $2p/3$. For $q = 7$ there are six root classes. Here $D - \Lambda$ is only about $4 \cdot 10^{-3}$, which explains the large exponent. A small further gain in either constant would lower it substantially (Section 11). The certificate for $q = 7$ therefore takes $\varepsilon_0 = 10^{-6}$ in the function Υ, where the rows $q = 2, 3, 4, 5$ take $\varepsilon_0 = 10^{-3}$. Its trial measure has 22 polynomial pieces, and its cells on Γ at the breakpoints of σ use the truncated logarithm of §6.4. For $q = 2$ there is one root class, $j = 1$, with first pole $k_p = \frac{p+1}{2}$.

The certificates of the two planes also take $\varepsilon_0 = 10^{-6}$. On the second plane $D - \Lambda$ is about 0.021, which explains the size of its exponent. On the first plane the points of Theorem 1.2(iii)–(v) gain only in Λ, since they use the certificate of the plane.

11. Limits of the method

The exponents are determined by three constants through $\rho/(D - \Lambda)$. Since $\partial\mu/\partial D = -\rho/\eta^2$, the exponent is most sensitive to the decay where η is small. For the shifts $\frac{1}{4}$ and $\frac{3}{4}$, $\rho/\eta^2 \approx 136$. For the shifts $\frac{k}{7}$ it is about $4.4 \cdot 10^5$.

On the arithmetic side, the profiles $\mathcal{P}_j$ are minima of the bounds of Theorem 4.5. Every further local bound lowers Λ by the integral of the amount by which it lies below $\mathcal{P}_j$. The denominator exponent of the method is the integral of the true exponent e_p, which the local bounds approximate from above. At third order this approximation is close to exhaustion for the shifts $\frac{k}{5}$ and $\frac{k}{7}$. We computed the Smith normal forms exactly at scales $r \le 2400$ and primes $p \le 2m$. There the minimum of the local bounds, with (U_n) included, equals the true exponent at all but a few isolated primes. For the shifts $\frac{k}{7}$ the invariants of valuation at least 4 contribute about $5 \cdot 10^{-6}$ to Λ. The bounds (U_n) gain about $7 \cdot 10^{-4}$. So at third order the local bounds are essentially exhausted. These computations are not part of the proof.

On the analytic side, Remark 6.4 explains why a certificate of the present form cannot do better than the constrained equilibrium measure on the chosen contour. A larger decay rate needs a different contour or a different treatment of the energy problem. The geometry parameter c trades the two

sides against each other. A longer border increases D, through the larger node measure. It also increases Λ, through the border rows and columns of the assignment bound.

12. Formalization

The theorems of Section 1 are formalized in Lean 4 [11] with Mathlib [41], together with every theorem, proposition and lemma their proofs use. The remarks and Section 11 are not part of the formalization. The development is available at

https://github.com/slitsey/catalan-irrationality-measure

It combines analytic and arithmetic proofs with generated rational data. Kernel checks verify the tables for Λ and the decay certificates. Numerical computations alone are not used as proof.

The statements are in the file `ChallengeHybrid.lean`. Their proofs are in `SolutionHybrid.lean`, in the namespace `TrigammaDifferences`. Each number ξ is written as a series: the classical form where there is one, and $\Psi(s,t)$ otherwise. Each theorem states that $b^{-\mu_0} < |\xi - a/b|$ for all $a \in \mathbb{Z}$ and all sufficiently large b. The plane theorems state this for all rationals a and b, not both zero, and the irrationality of the number. The theorems are the following.

- `plane1_measure`: Theorem 1.1, with $\mu_0 = 20.6622$.
- `chi3_measure`: Theorem 1.2(i), with $\mu_0 = 13.7046$.
- `chi3_plus_pi2_measure` and `chi3_minus_pi2_measure`: Theorem 1.2(ii), with $\mu_0 = 9.4998$.
- `dip_2_27_plus_measure`, `dip_2_27_minus_measure`, `dip_1_27_plus_measure`, `dip_1_27_minus_measure`, `dip_1_18_plus_measure` and `dip_1_18_minus_measure`: Theorem 1.2(iii)–(v), with $\mu_0 = 14.000$, 15.359 and 18.161.
- `plane2_measure`: Theorem 1.3, with $\mu_0 = 335.20$.
- `zeta3_measure` and `pi2_measure`: Theorem 1.4, with $\mu_0 = 6.0382$ and 6.5253.
- `lambda_plus_measure` and `lambda_minus_measure`: Theorem 1.5(i), with $\mu_0 = 31.2876$.
- `fifths1_measure`, ..., `fifths4_measure`: Theorem 1.5(ii), with $\mu_0 = 23.8272$.
- `sevenths1_measure`, ..., `sevenths6_measure`: Theorem 1.5(iii), with $\mu_0 = 1806.1719$.

The shared results are in the namespace `CatalanCore.Uniform`.

- `Content.BankShared.Check.hasScaledContent_of_check` is Theorem 4.14. The local bounds of Theorem 4.5 are proved in `Check/Catalogue.lean`, `Check/LP2W.lean`, `Check/Prefix.lean` and `Check/UF.lean`, with the closed rank bound and the branches of Theorem 4.8. Its form for functions with several first poles is `Content.BankShared.KTerm.hasScaledContent_of_kcheck`.

- `Content.BankShared.TwoTerm.LP2HypW.gle` is Theorem 4.6.
- `Content.BankShared.TwoTerm.Pre.prefix_gle` is Theorem 4.7.
- `Content.UFSimple.WrapCheck.catValid_wrap` and `Content.UFSimple.WrapPrefixCheck.catValid_prefix_full` are Theorem 4.8 for $n = 0$ and for $n \geq 1$. Lemma 4.11 is `Content.UFSimple.WrapPrefixClosed.HSplit.det_ne` with `HSplit.dir`.
- `BlockGeom.hasNV0` is Theorem 5.1.
- `Decay.AltDecay.altGeom_hasScaledDecay` is Theorem 6.2. It uses the node energy check `Decay/Nodes/EnergyCheck.lean`, the contour check `Decay/ContourCheck.lean` and the cell checks `Decay/PolyWire3`. For the shifts $\frac{k}{7}$ it is applied in the form `Decay.AltDecay.altGeom_hasScaledDecay_Tr`, with the same conclusion. There the cells on the contour at breakpoints use the truncated logarithm, proved in `Decay/Breakpoint/TruncLog.lean` and `Decay/Breakpoint/CellT.lean`.
- `measure_of_family_nv0_even` is Theorem 10.1 at the even scales.

The solution has no `sorry` and no `native_decide`. The command `#print axioms` reports only the three standard axioms `propext`, `Classical.choice` and `Quot.sound`. The prime number theorem in arithmetic progressions enters the asymptotics of the denominators. Its formal source is the PNT+ project [42], through OpenAI's workspace and compatibility additions. We use its Wiener–Ikehara theorem.

The proofs were checked with Lean `v4.35.0-rc2` and the matching Mathlib release. The vendored files from OpenAI's repository and from the PNT+ project were converted to Lean's module system and ported to the current toolchain. The repository's README and provenance documents describe the changes. The certificate data were generated with the interval arithmetic of Arb [19], through `python-flint`, and Lean re-checks them.

Disclosure of AI use

The mathematics, the Lean formalization, the certificate generators and the text of this paper were produced by AI systems at the request of the author. The author posed the problems, directed the work and is responsible for the paper. The systems were Claude Opus 5.5, Claude Sonnet 5.5 and Claude Haiku 5.5 (Anthropic), running in Claude Code, and GPT-6.1 Sol (OpenAI), running in Codex. The underlying irrationality proof for Catalan's constant and its Lean formalization are due to OpenAI [32].

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